

Time Frequency Analysis Overview

- Introduction and motivation
- Windowed estimates as filter banks
- Uncertainty principle
- Examples
- Other estimates

Introduction and Motivation

$$r_x(\ell, n) = \mathbb{E} [x(n)x(n - \ell)^*] \quad \text{for } m \leq \ell \leq m$$

$$\begin{aligned} R_x(e^{j\omega}, n) &= \sum_{\ell=-\infty}^{\infty} r_x(\ell, n) \\ &\approx \sum_{\ell=-m}^m r_x(\ell, n) \end{aligned}$$

- In practical applications, many signals are nonstationary
- In order to apply the techniques, we usually assume the signal statistics change “slowly”
- “Locally” stationary
- Rate of change depends on the process
- Time frequency analysis tries to estimate how these properties change with time

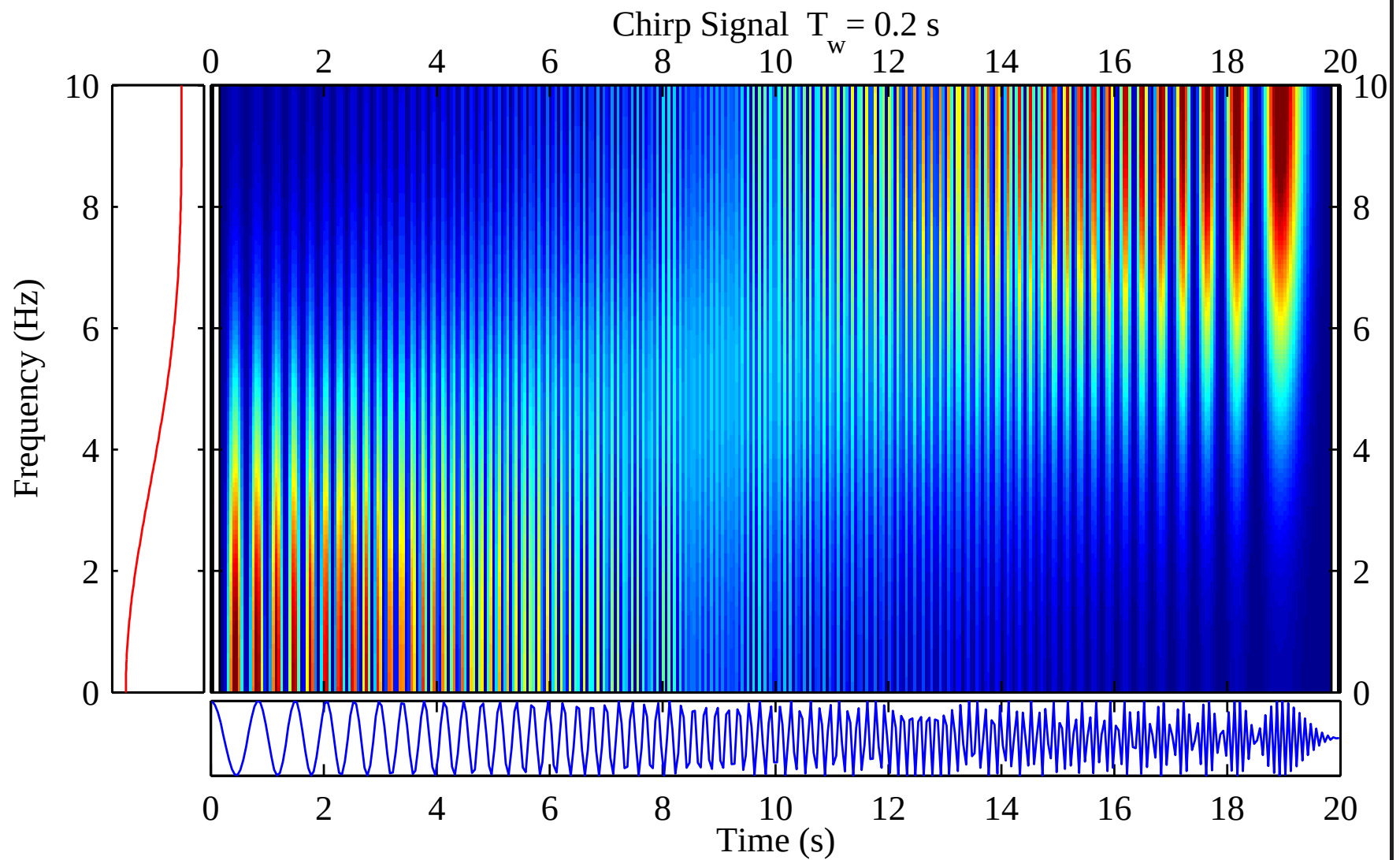
Practical Application

- In practice, time-frequency analysis is most frequently performed by estimating the statistical properties from sliding windows
- Resulting estimates, e.g., $R(e^{j\omega}, n)$, are a function of two independent variables
- Typically represented with three-dimensional meshes, contour plots, or pseudo-colored images
- When used to estimate the time-varying PSD, are called **spectrograms** or **short-time Fourier transforms (STFT)**

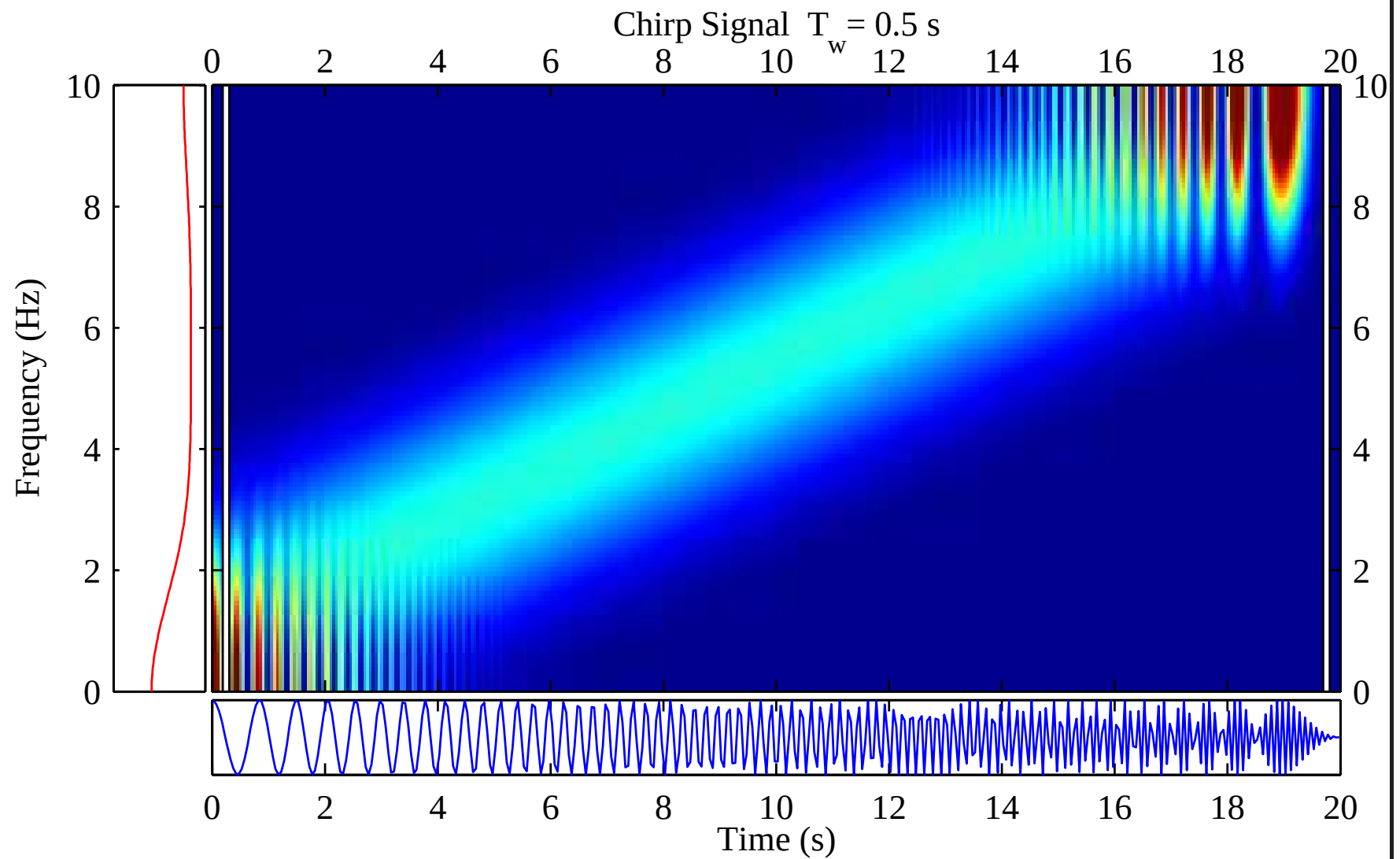
Example 1: Chirp Spectrogram

Apply the modified periodogram on a chirp signal with a linear frequency sweep from 1–10 Hz over a period of 20 s.

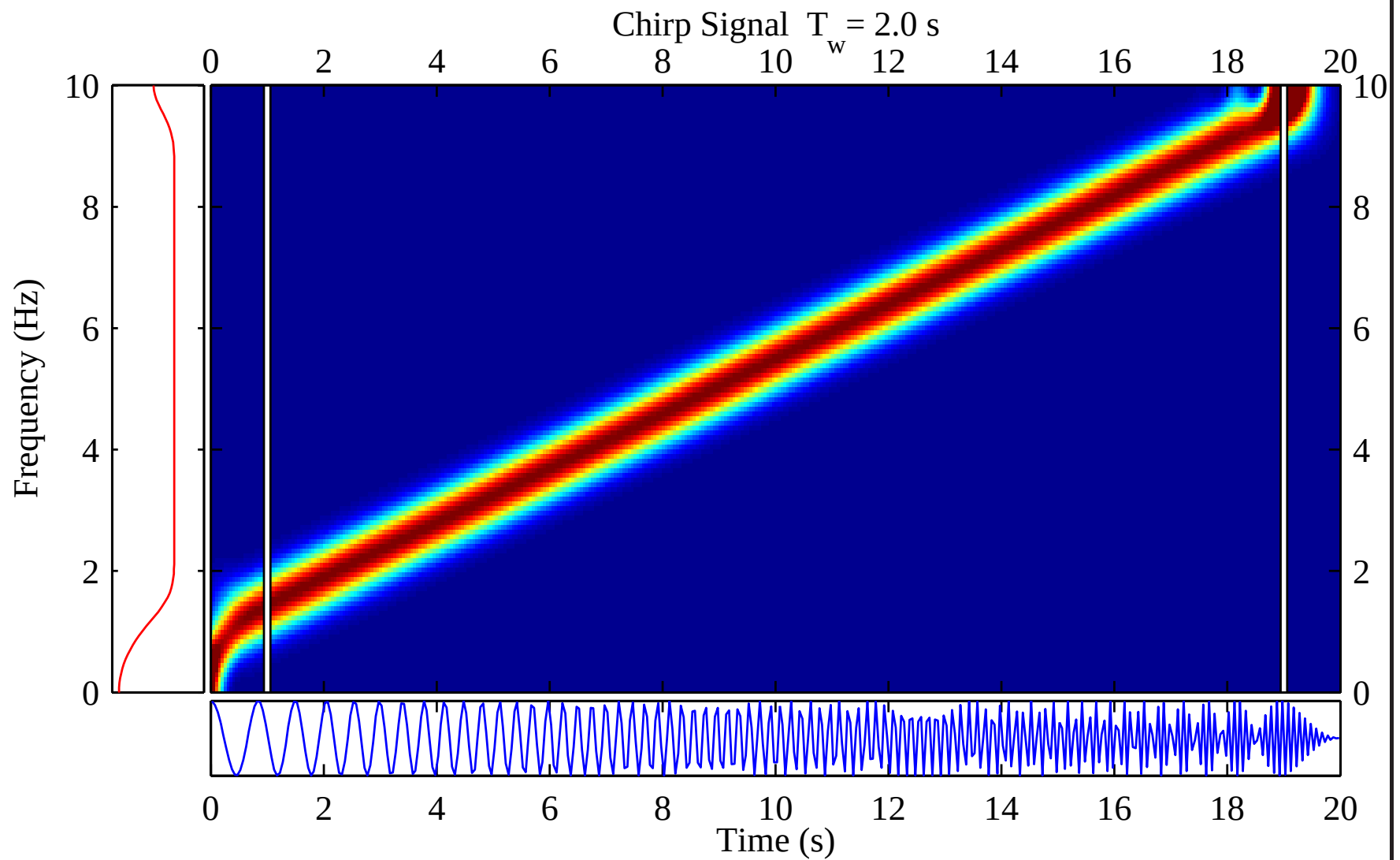
Example 1: Chirp Spectrogram



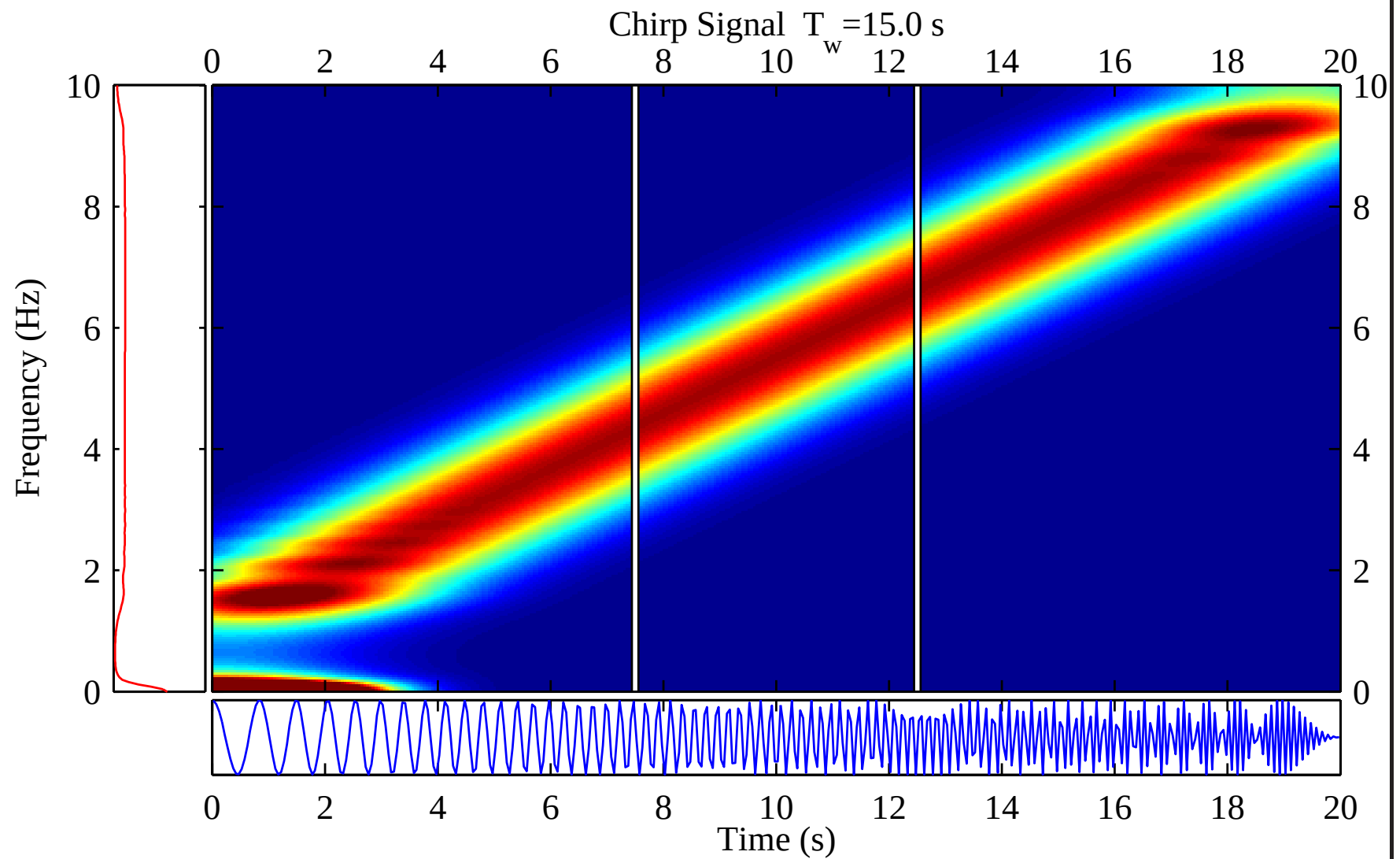
Example 1: Chirp Spectrogram



Example 1: Chirp Spectrogram



Example 1: Chirp Spectrogram



Example 1: Relevant MATLAB Code

```
clear;
close all;

fs = 20;           % Sample rate (Hz)
T = 20;           % Signal duration (s)
f = [1 10];       % Chirp frequencies [start stop]
Tw = [0.2 0.5 1 2 5 10 15]; % Window lengths

k = 1:T*fs;       % Discrete time index
t = (k-0.5)/fs;   % Sample times (s)

for c0=1:6,
    switch c0,
        case 1,
            x = chirp(t,f(1),T,f(2));
            sg = 'Chirp';
        case 2,
            x = cos(2*pi*5*t);
            sg = 'Cosine';
        case 3,
            [jnk,id] = min(abs(t-T/2));
            x = zeros(length(k),1);
            x(id) = 1;
            sg = 'Impulse';
        case 4,
            id = find(t>T/2);
            x = zeros(length(k),1);
```

```

        x(id) = 1;
        sg = 'UnitStep';
    case 5,
        id = 1:1*fs:max(k);
        x = zeros(length(k),1);
        x(id) = 1;
        sg = 'ImpulseTrain';
    case 6,
        x = randn(length(k),1);
        sg = 'WhiteNoise';
    end;
for c1=1:length(Tw),
    tw = Tw(c1);
    [S,ts,f] = NonparametricSpectrogram(x,fs,tw);
    figure;
    FigureSet(1,'LTX');
    I = abs(S).^2;
    cr = prctile(reshape(I,prod(size(I)),1),[0.5 99.5]);
    ha = axes('Position',[0.14 0.20 0.80 0.73]);
    h = imagesc(ts,f,abs(S).^2,cr);
    hold on;
        h = plot(tw/2*[1 1],[0 fs/2],'k',tw/2*[1 1],[0 fs/2],'w');
        set(h(1),'LineWidth',2);
        set(h(2),'LineWidth',1);
        h = plot((T-tw/2)*[1 1],[0 fs/2],'k',(T-tw/2)*[1 1],[0 fs/2],'w');
        set(h(1),'LineWidth',2);
        set(h(2),'LineWidth',1);
    hold off;
    set(gca,'YDir','Normal');
    set(gca,'XAxisLocation','Top');
    set(gca,'YAxisLocation','Right');

```

```

        xlim([0 T]);
        ylim([0 fs/2]);
        ht = title(sprintf('%s Signal   T_w=%4.1f s',sg,tw));

        ha = axes('Position',[0.14 0.10 0.80 0.09]);
        h = plot((k-0.5)/fs,x,'b');
        set(h,'LineWidth',0.5);
        xlim([0 T]);
        set(ha,'YTick',[])
        xlabel('Time (s)');

        ha = axes('Position',[0.07 0.20 0.065 0.73]);
        Rx = mean(I,2);
        h = plot(Rx,f,'r');
        set(h,'LineWidth',0.5);
        ylim([0 fs/2]);
        set(ha,'XTick',[])
        ylabel('Frequency (Hz)');

        AxisSet(8);
        set(ht,'VerticalAlignment','Top')
        print(sprintf('%sEx%03d',sg,tw*10),'-depsc');
    end;
end;

```

Why the Windowed Periodogram?

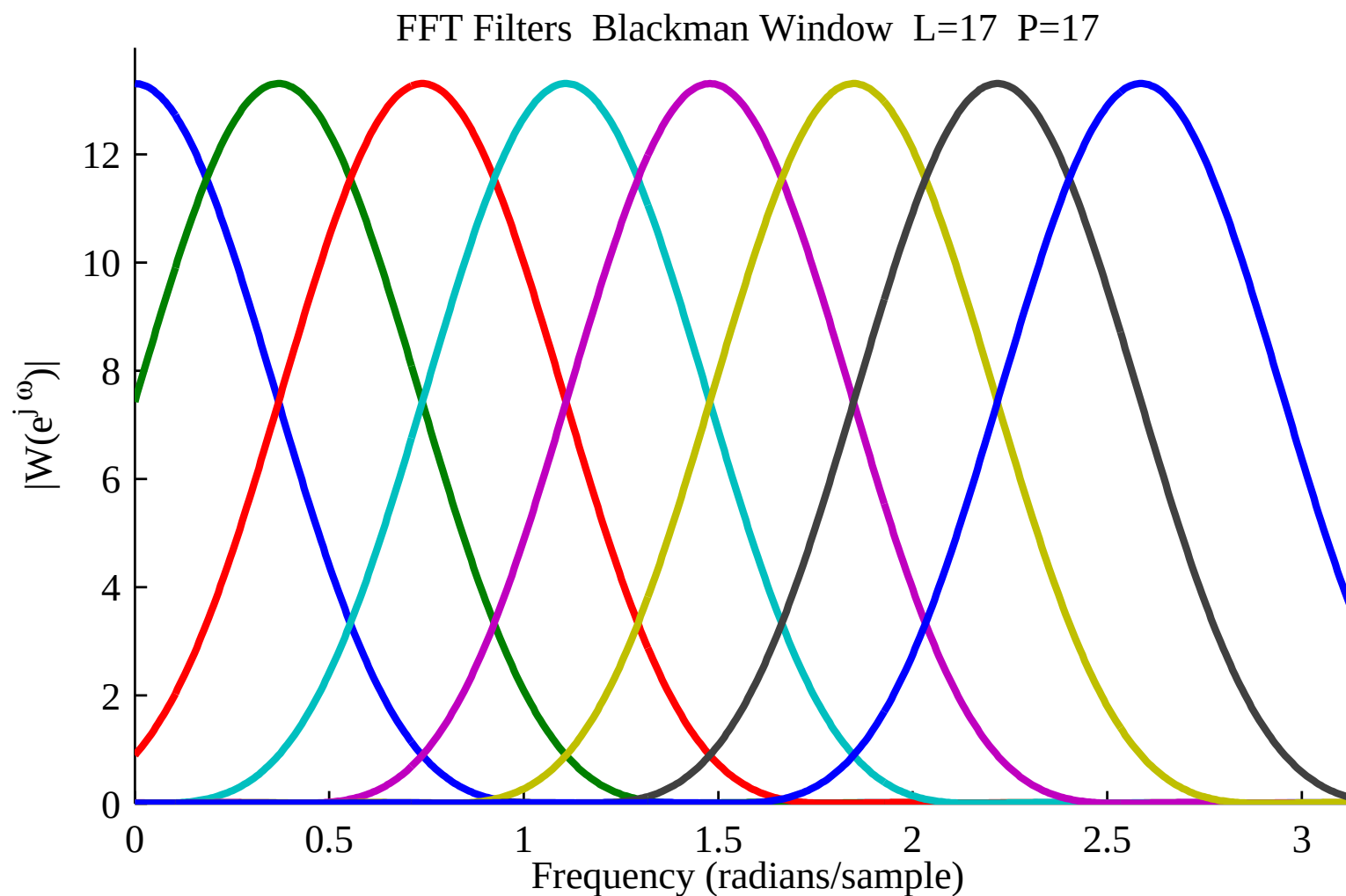
- Why not a smoother technique with less variance?
 - The eye can smooth the plots visually
 - Techniques to decrease variance decrease the resolution too much
- Why window the segments? Could use a rectangular window
 - If tapered window is not applied, estimate will vary even if the signal is a constant sinusoid
 - Due to variation of phase of sinusoid that is included in the segment (covered by the window)
 - Taper reduces this effect
 - Also reduces sidelobe leakage

Filter Bank Interpretation

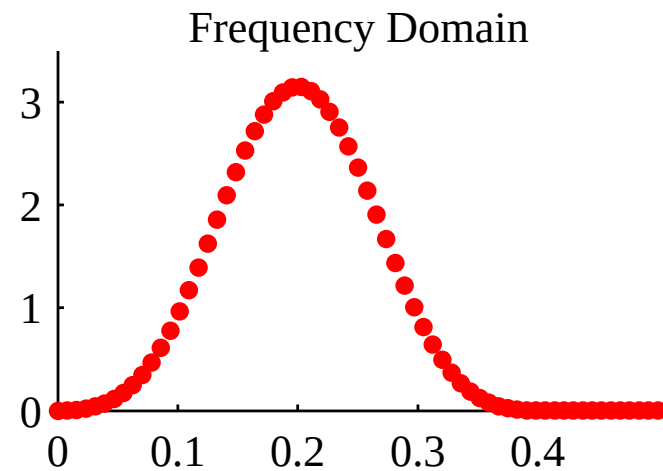
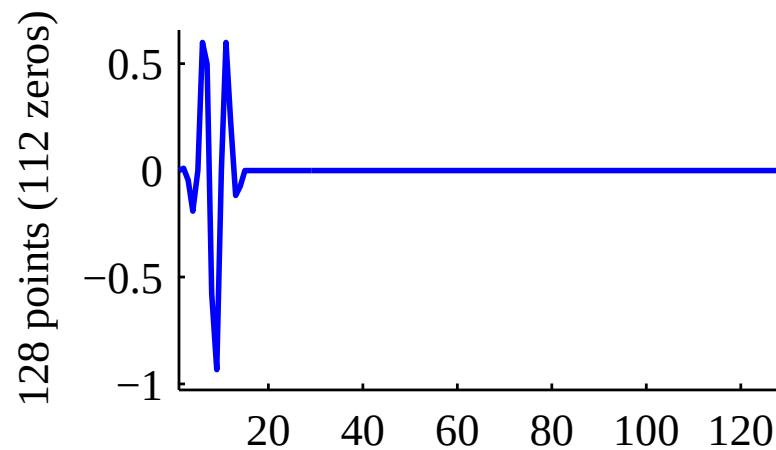
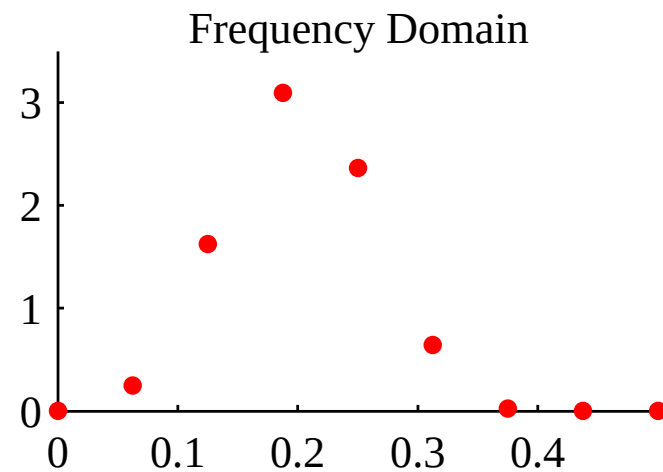
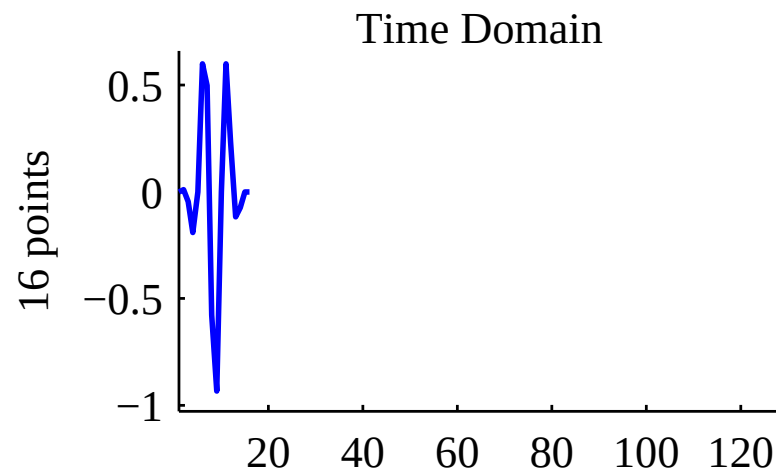
$$\begin{aligned} X(e^{j\omega}, n) &= \sum_{\ell=-\infty}^{\infty} x(n + \ell)w(\ell)e^{-j\omega\ell} \\ &= \sum_{m=-\infty}^{\infty} x(m)w(m - n)e^{-j\omega(m-n)} \\ &= \sum_{m=-\infty}^{\infty} x(m)w(-(n - m))e^{j\omega(n-m)} \\ &= x(n) * w(-n)e^{j\omega n} \\ \hat{R}^{(\text{MP})}(e^{j\omega}, n) &= \frac{1}{L}|X(e^{j\omega}, n)|^2 \end{aligned}$$

- The spectrogram can be thought of as the squared magnitude of the output of an LTI system with impulse response $h_{\omega}(n) = w(-n)e^{j\omega n}$
- The magnitude response evaluated at ω_o is then $W(e^{j(\omega_o - \omega)})$

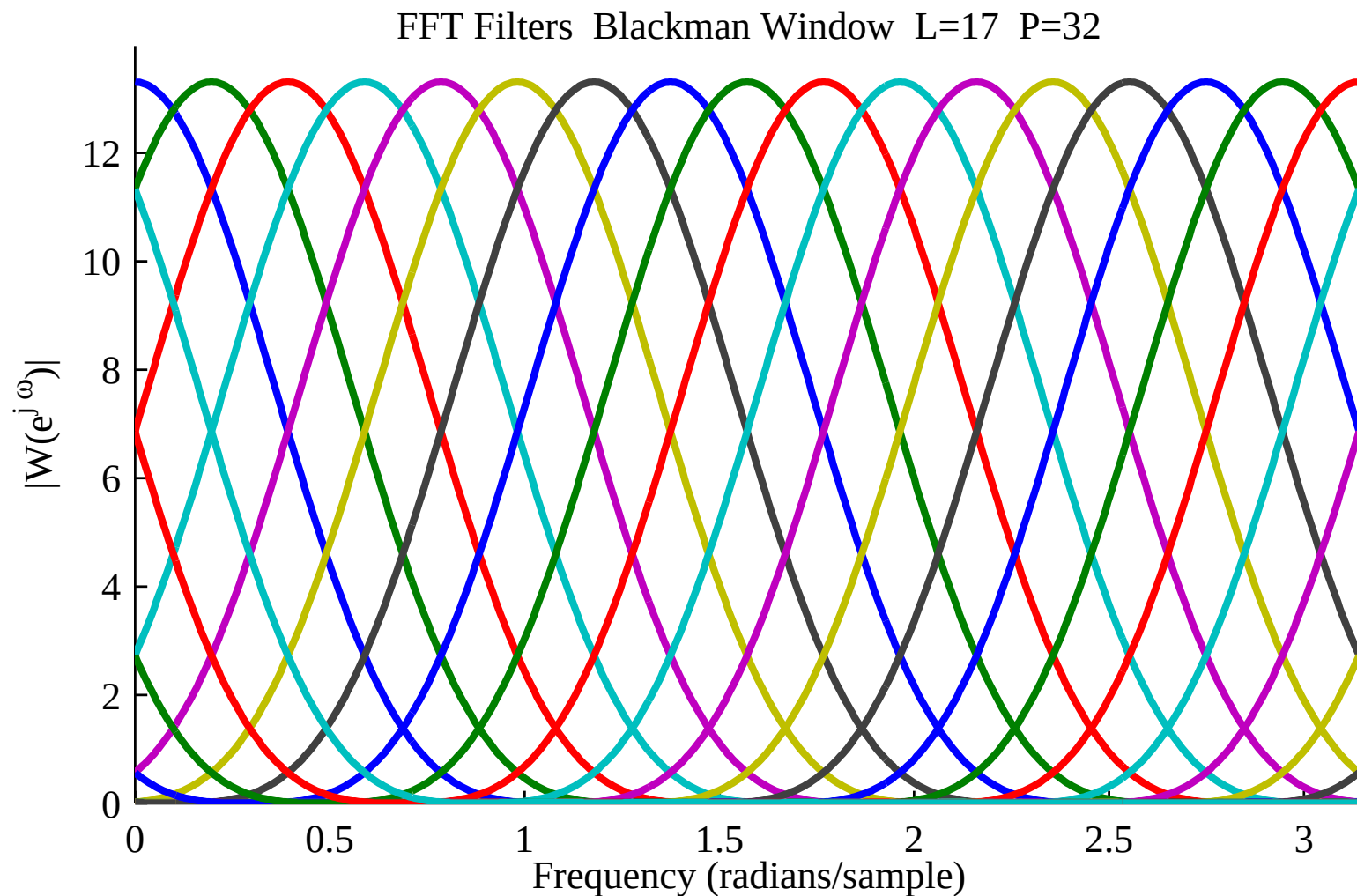
Filter Bank Frequency Responses without Padding



Effect of Zero Padding



Filter Bank Frequency Responses with Padding



Primary Tradeoff: Segment Length

- Primary purpose of windowing is to guarantee local stationarity
- Window shape controls tradeoff between sidelobe leakage and main lobe width
- Primary parameter is the window length, L
- Long window
 - Poor time resolution
 - Good frequency resolution
 - Use with slowly varying signal characteristics
- Short window
 - Good time resolution
 - Poor frequency resolution
 - Use with rapidly changing signal characteristics
- Thus L primarily controls the tradeoff between time and frequency resolution

Energy Density Functions

- Note that the windowed signal segments have finite energy
- Analogous ideas apply in continuous time
- Consider, for a moment, a continuous time windowed segment $x(t)$
- Defined the normalized **temporal energy density function** and **spectral energy density function** as

$$e_x(t) = \frac{|x(t)|^2}{\int_{-\infty}^{\infty} |x(t)|^2 dt} \quad E_x(\omega) = \frac{|X(j\omega)|^2}{\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega}$$

- Like pdfs, both of these functions are nonnegative and have unit area
- Can be thought of as energy density functions (edfs) in time and frequency
- Like pdfs, we can define **energy moments**

Bandwidth

$$\mu_{t,x} = \int_{-\infty}^{\infty} t e_x(t) dt$$

$$\mu_{\omega,x} = \int_{-\infty}^{\infty} \omega E_x(\omega) d\omega$$

$$\sigma_{t,x}^2 = \int_{-\infty}^{\infty} |t - \mu_{t,x}|^2 e_x(t) dt \quad \sigma_{\omega,x}^2 = \int_{-\infty}^{\infty} |\omega - \mu_{\omega,x}|^2 E_x(\omega) d\omega$$

- There are many definitions of *duration* and *bandwidth* of a signal
- One pair of possible definitions is based on the second-order moments of the energy density functions
- Let us define the signal
 - **Duration** as the standard deviation of the temporal edf, $\sigma_{t,x}$
 - **Bandwidth** as the standard deviation of the spectral edf, $\sigma_{\omega,x}$

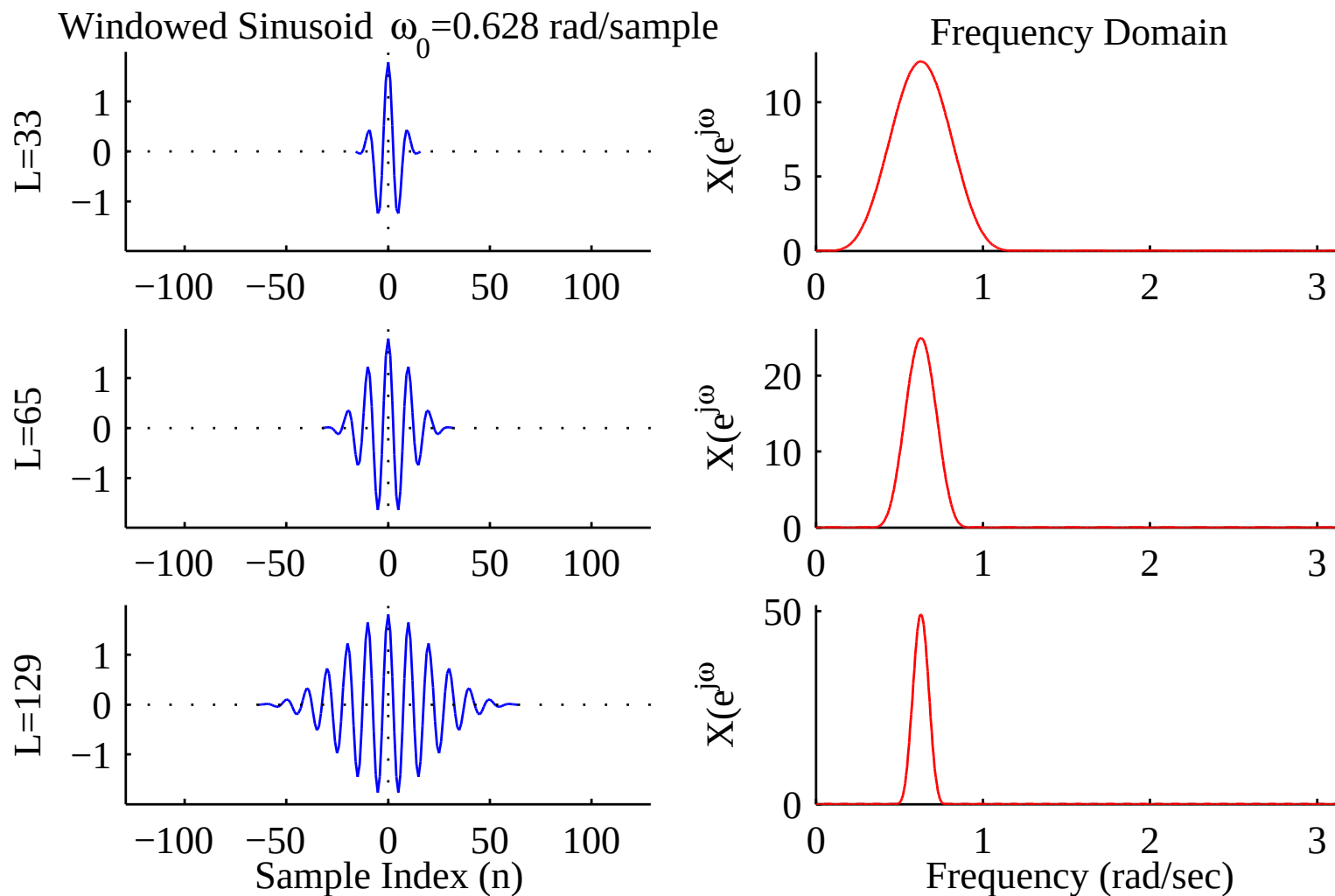
Uncertainty Principle

It can be shown that

$$\sigma_{t,x} \times \sigma_{\omega,x} \geq \frac{1}{2}$$

- In this context, it is not actually an expression of uncertainty
 - Signals with short duration have broad bandwidth
 - Signals with long duration can have narrow bandwidth
- Equivalently, if one density is narrow then the other is wide
- They can't both be arbitrarily narrow
- In quantum mechanics, applies to probability densities of position and velocity so *uncertainty* is appropriate
- Not aware of an equivalent uncertainty principle for DT signals

Time Bandwidth Product



MATLAB Code

```
%function [] = TimeBandwidthEx();
close all;

NZ = 2^10;
w0 = pi/5;
L = [32 64 128]+1;

nL = length(L);
k = (0:NZ-1)';
w = k*2*pi/NZ;
kw = 1:NZ/2;

l = 1001;
wn = blackman(l);
wn = wn*sqrt(1/sum(wn.^2));
n = (-(l-1)/2:(l-1)/2)';
x = cos(w0*n).*wn;
ymx = max(x);
ymn = min(x);
yrg = ymx-ymn;

figure;
FigureSet(1,'LTX');

for c1=1:nL,
    l = L(c1);
    n = (-(l-1)/2:(l-1)/2)';
```

```

wn = blackman(1+2);
wn = wn(2:end-1);           % Trim off the zeros
wn = wn*sqrt(1/sum(wn.^2)); % Scale appropriately
x = cos(w0*n).*wn;          % Windowed signal segment
X = exp(j*(1-1)/2*w).*fft(x,NZ); % Calculate frequency response
%disp(max(imag(X)));
X = real(X);                 % Eliminate imaginary part due to finite resolution
subplot(nL,2,c1*2-1);
    h = plot(n,x);
    xlim([-max(L) max(L)]);
    ylim([ymn-0.05*yrg ymx+0.05*yrg]);
    AxisLines;
    box off;
    ylabel(sprintf('L=%d',1));
    if c1==1,
        title(sprintf('Windowed Sinusoid  \omega_0=%5.3f rad/sample',w0));
    elseif c1==nL,
        xlabel('Sample Index (n)');
    end;
subplot(nL,2,c1*2);
    h = plot(w(kw),X(kw),'r');
    xlim([0 pi]);
    ylim([0 1.05*max(X)]);
    AxisLines;
    ylabel('X(e^{j\omega})');
    box off;
    if c1==1,
        title('Frequency Domain');
    elseif c1==nL,
        xlabel('Frequency (rad/sec)');
    end;

```

```
    end;  
AxisSet(8);  
print('TimeBandwidthEx','-depsc');
```

Attaining the Uncertainty Principle

The Gaussian function attains the uncertainty principle with equality

$$x(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{t^2}{2\sigma^2}}$$

$$X(j\omega) = e^{-\frac{\omega^2\sigma^2}{2}}$$

$$e_x(t) = \frac{1}{\sqrt{\pi\sigma}} e^{-\frac{t^2}{\sigma^2}}$$

$$E_x(\omega) = \frac{1}{\sqrt{\pi\sigma^{-1}}} e^{-\omega^2\sigma^2}$$

$$\sigma_{t,x} = \frac{\sigma}{\sqrt{2}}$$

$$\sigma_{\omega,x} = \frac{1}{\sqrt{2}\sigma}$$

$$\sigma_{t,x}\sigma_{\omega,x} = \frac{1}{2}$$

- If this has the best duration-bandwidth product (i.e., resolution), why don't we use this as a window?

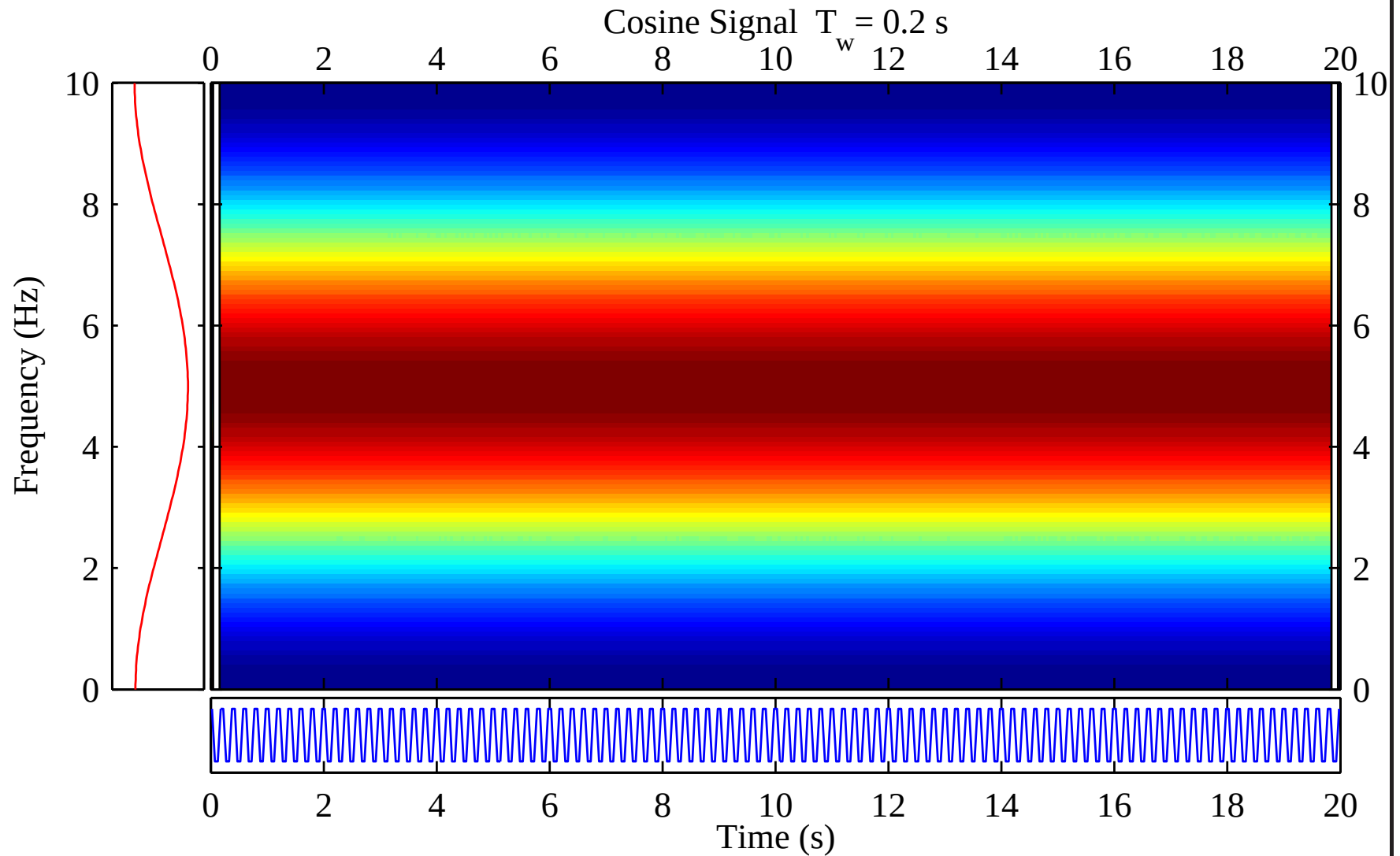
Practical Implementation of Spectrograms

- Remove the signal mean (center the signal)
 - Apply a highpass filter, if necessary
- The MATLAB implementation is not very good
- Time and frequency resolution should not exceed that of the display
- Use sufficient zero padding and time resolution
 - Make user-specified parameters
 - Do not ask for degree of overlap - calculate it
- Edge effects
 - Notify user of the extent of edge effects
 - Handle edges elegantly (repeat last value, extrapolate by prediction, etc.)
- Often easier to visualize the square root of the PSD, rather than the PSD

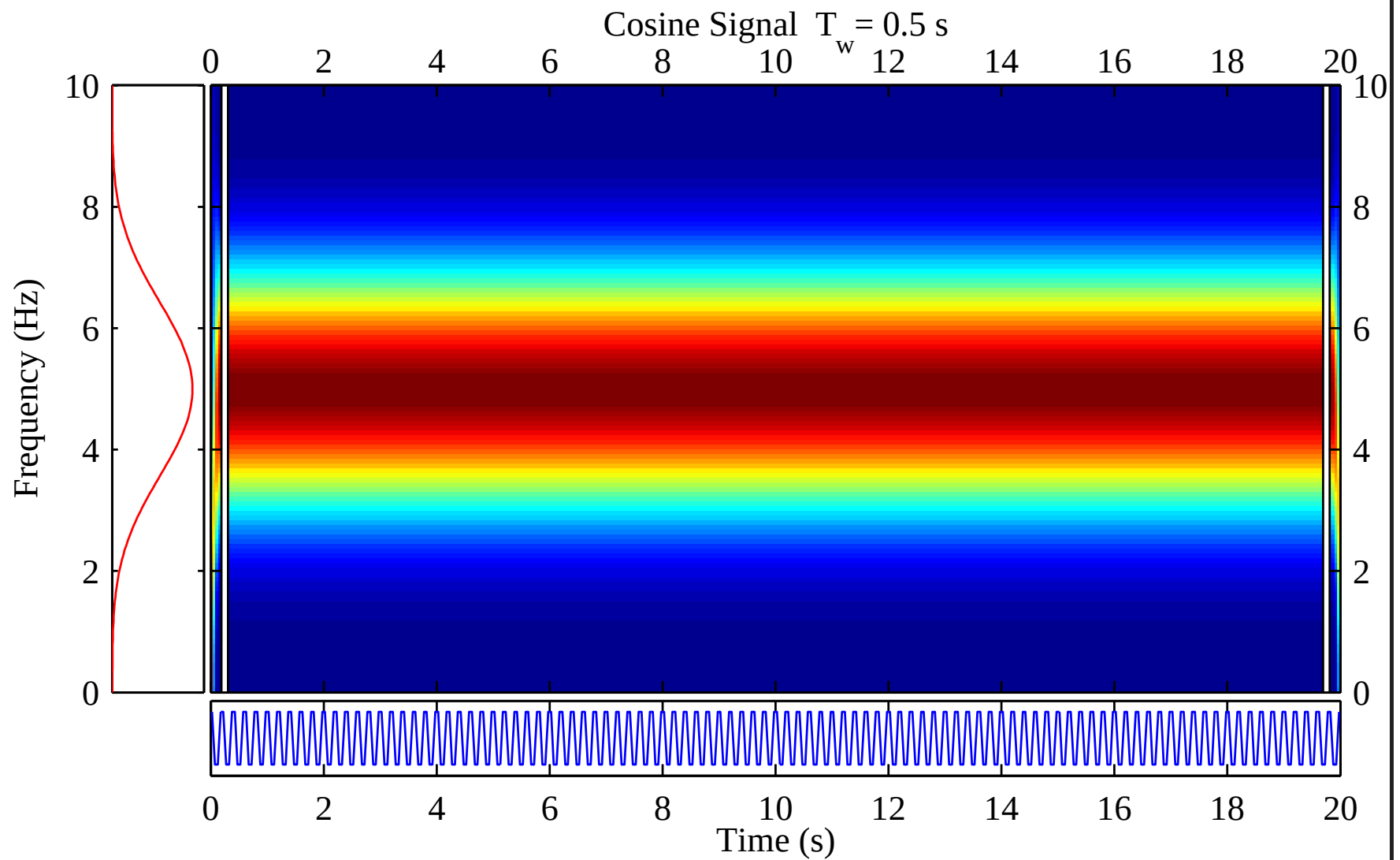
Example 2: Cosine, Impulse, and Step Spectrograms

Repeat the previous example for a 5 Hz sinusoidal signal, the unit impulse at $t = 10$ s, and the unit step starting at $t = 10$ s.

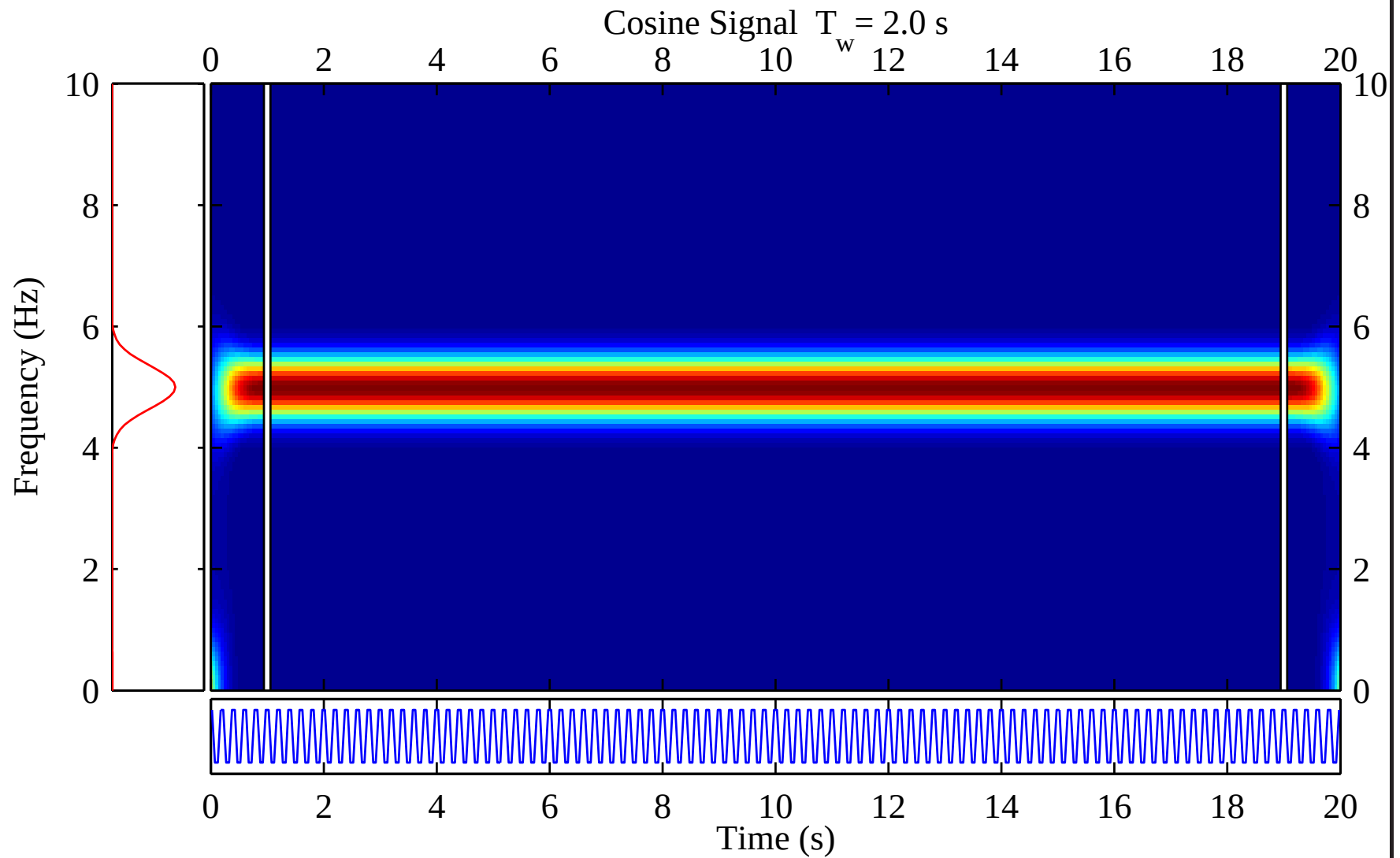
Example 2: Cosine Spectrogram



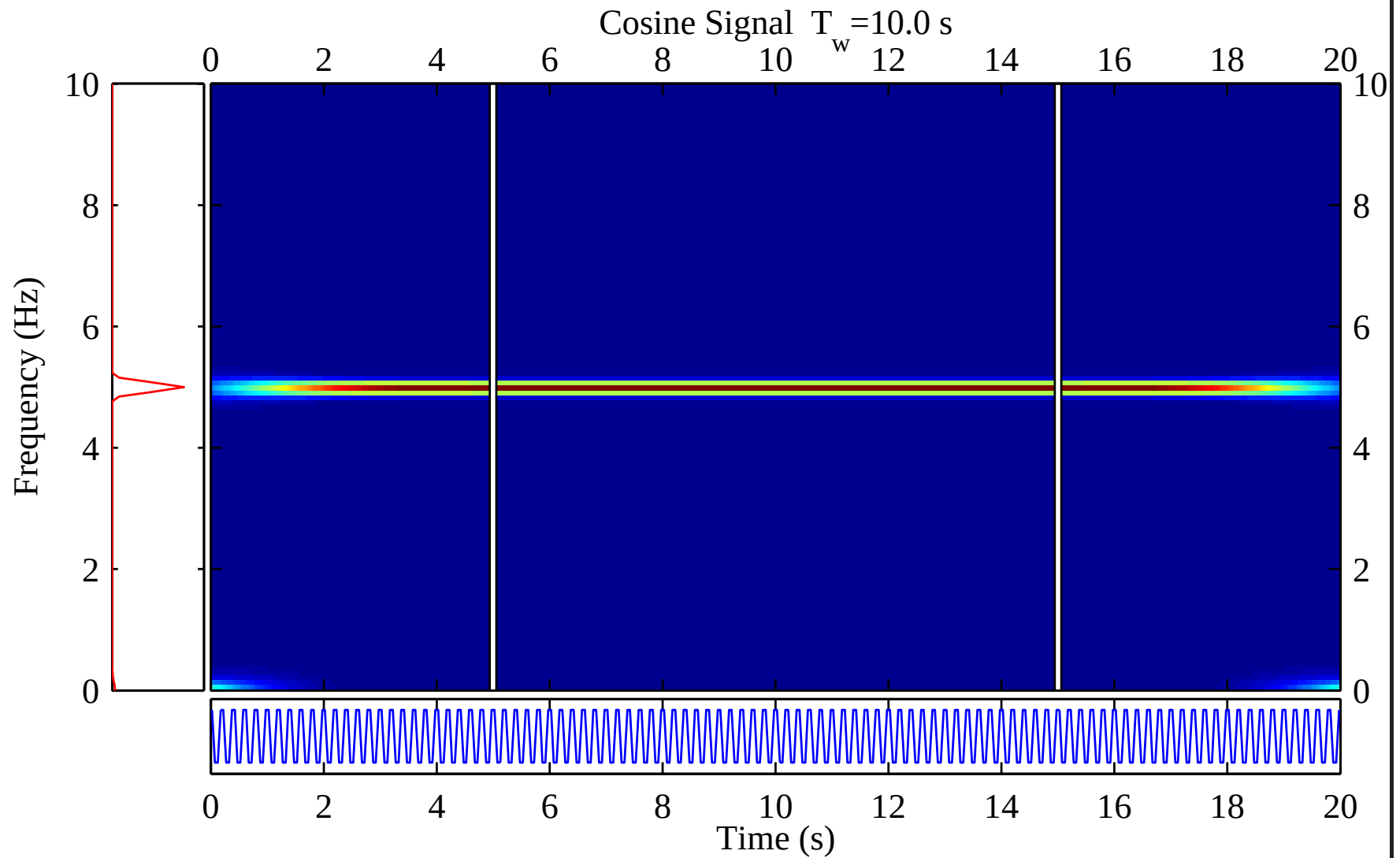
Example 2: Cosine Spectrogram



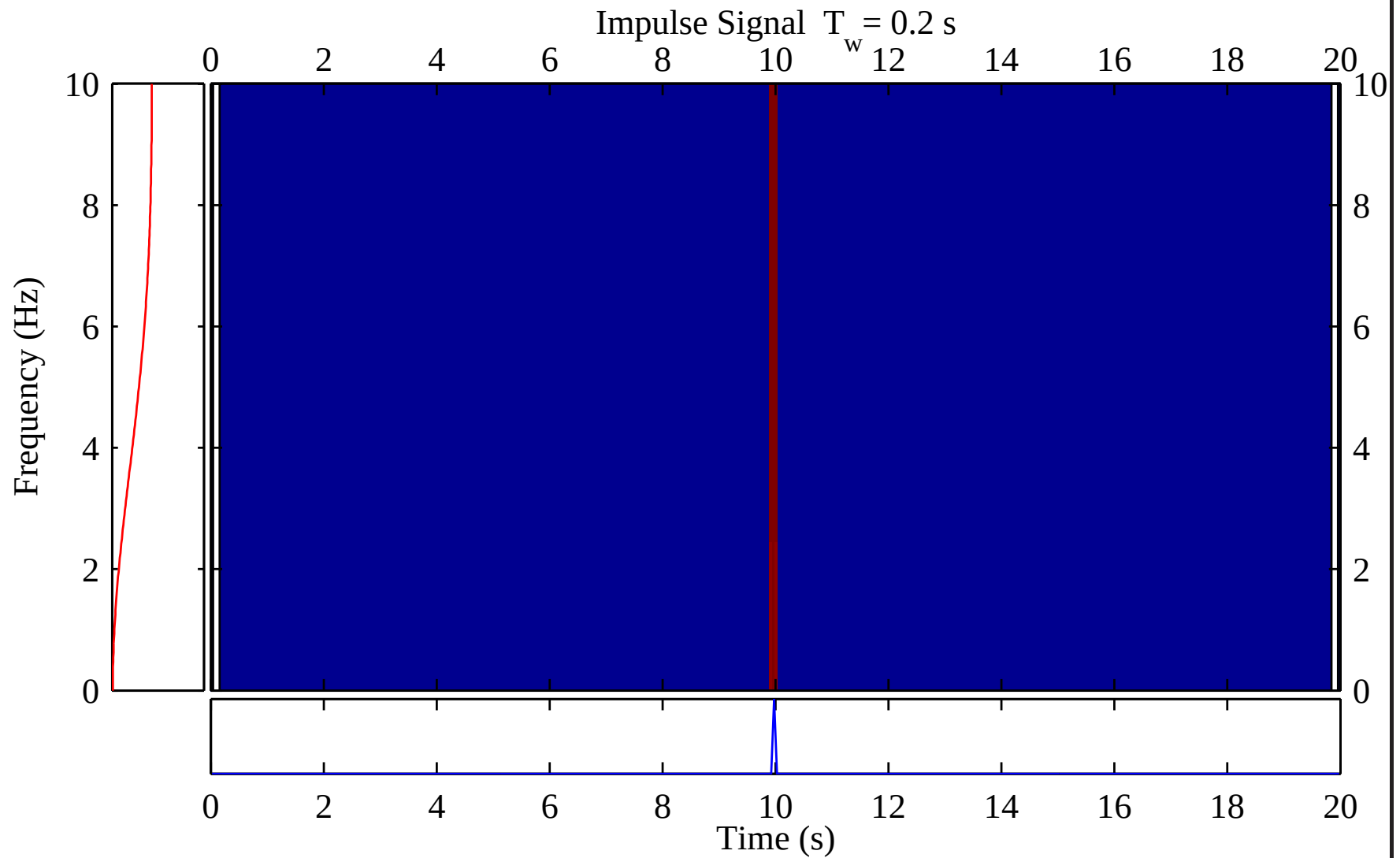
Example 2: Cosine Spectrogram



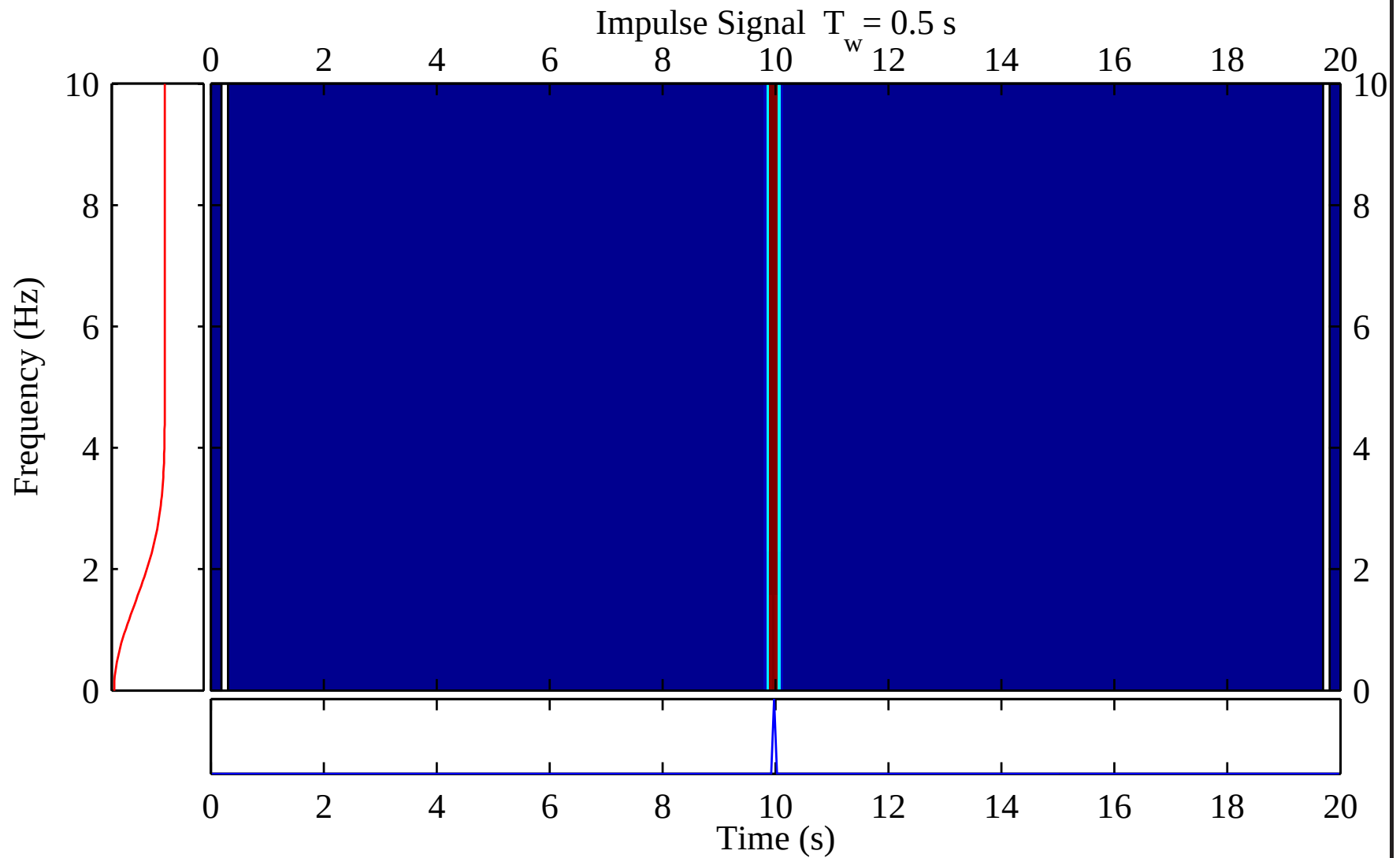
Example 2: Cosine Spectrogram



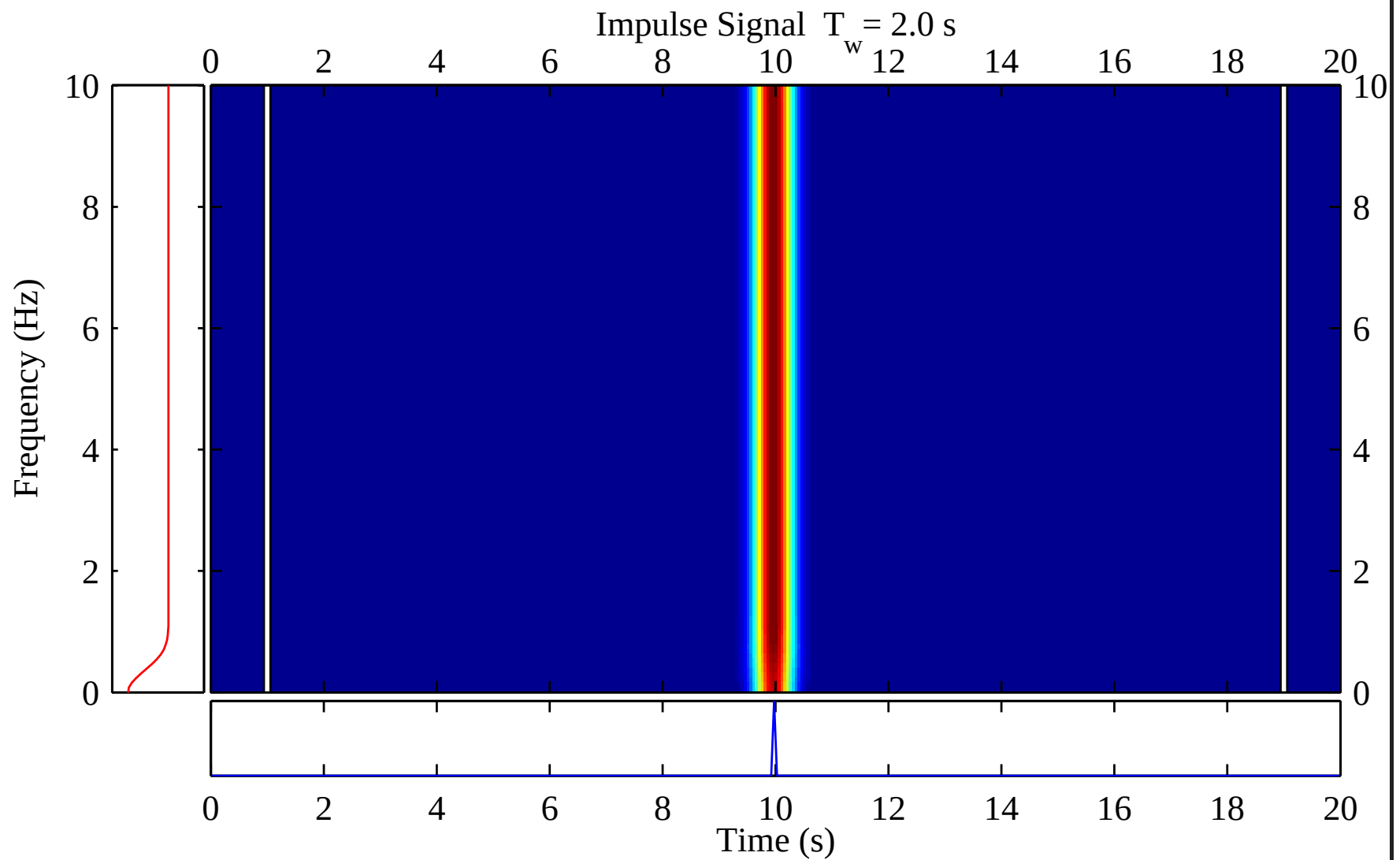
Example 2: Impulse Spectrogram



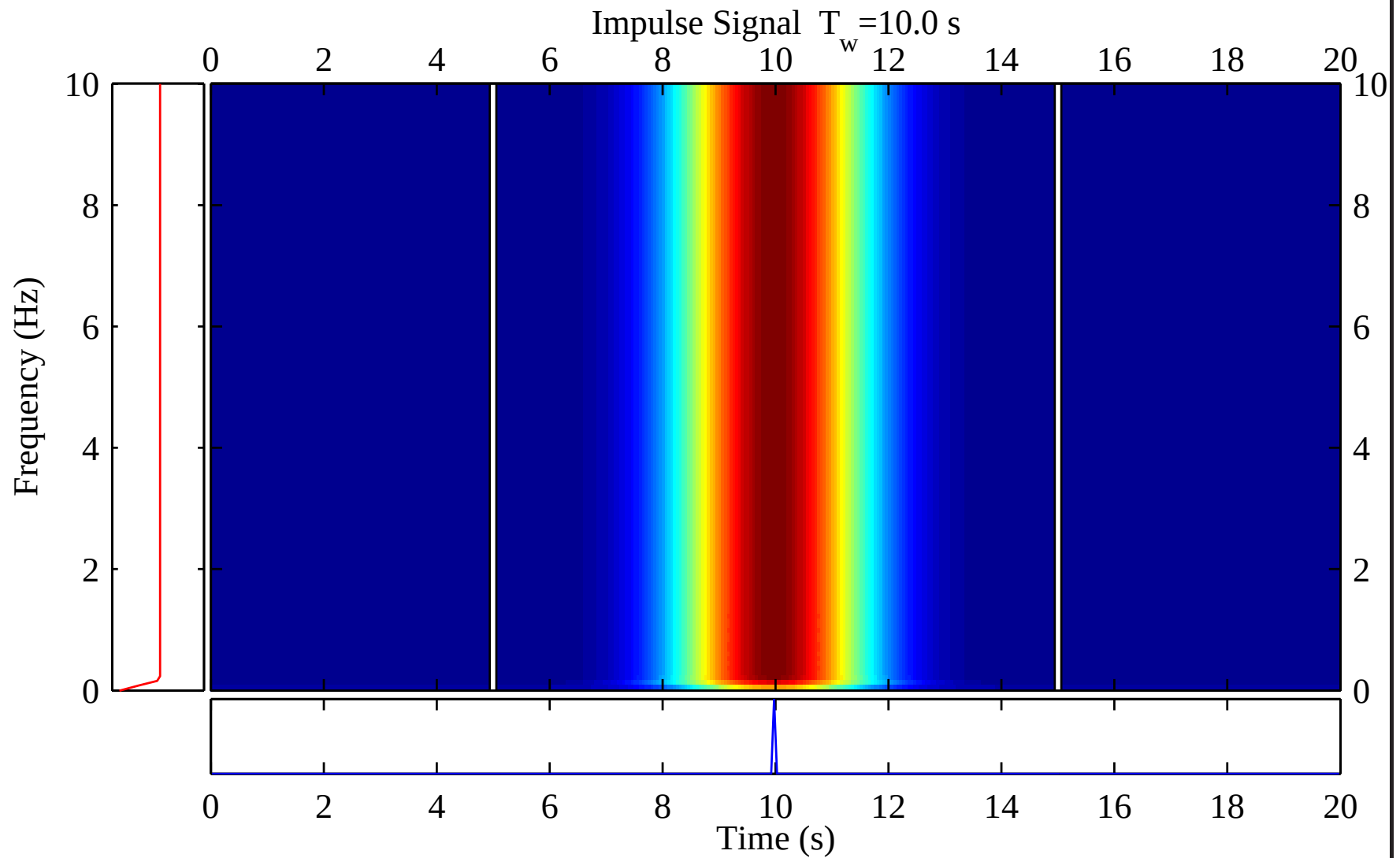
Example 2: Impulse Spectrogram



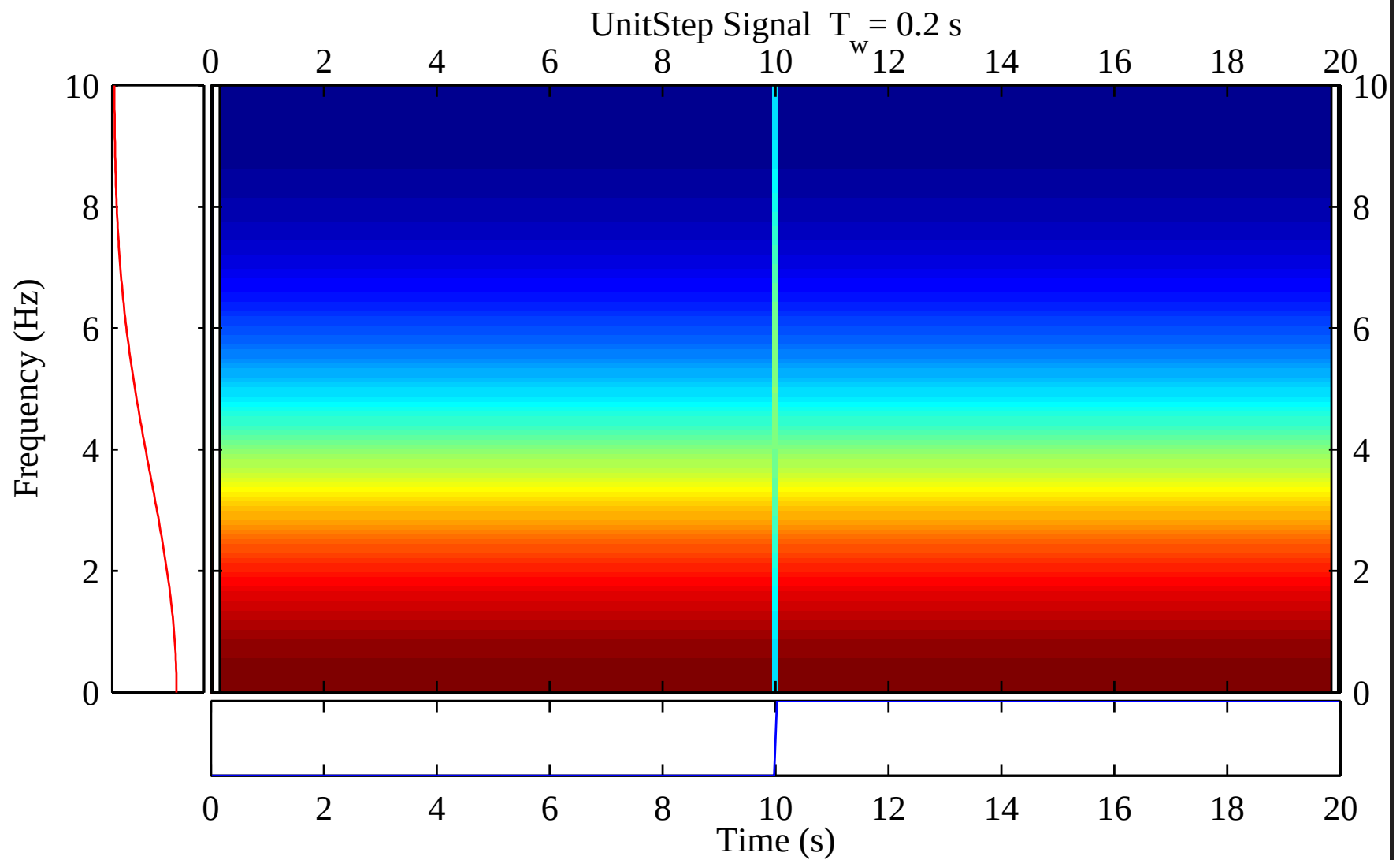
Example 2: Impulse Spectrogram



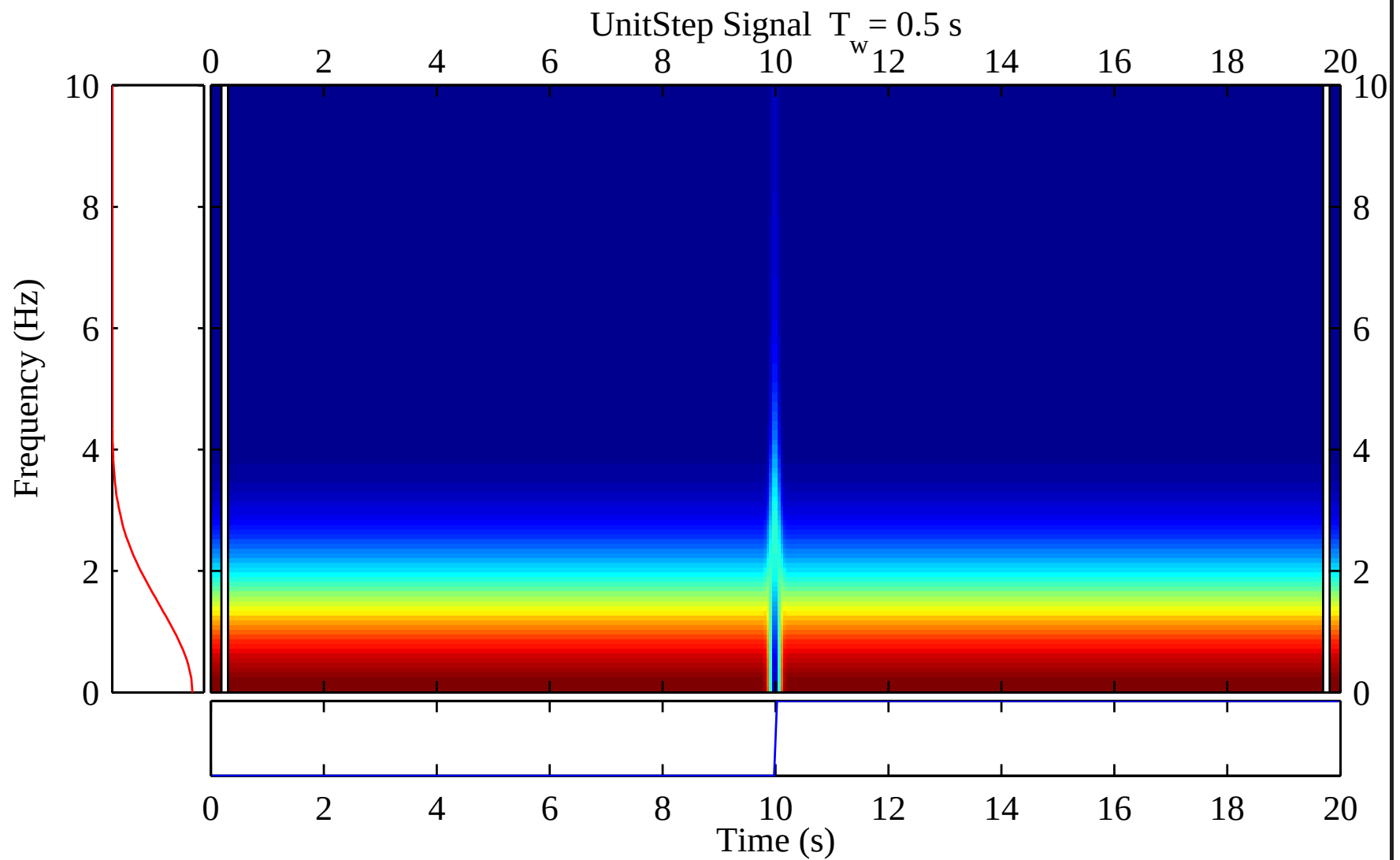
Example 2: Impulse Spectrogram



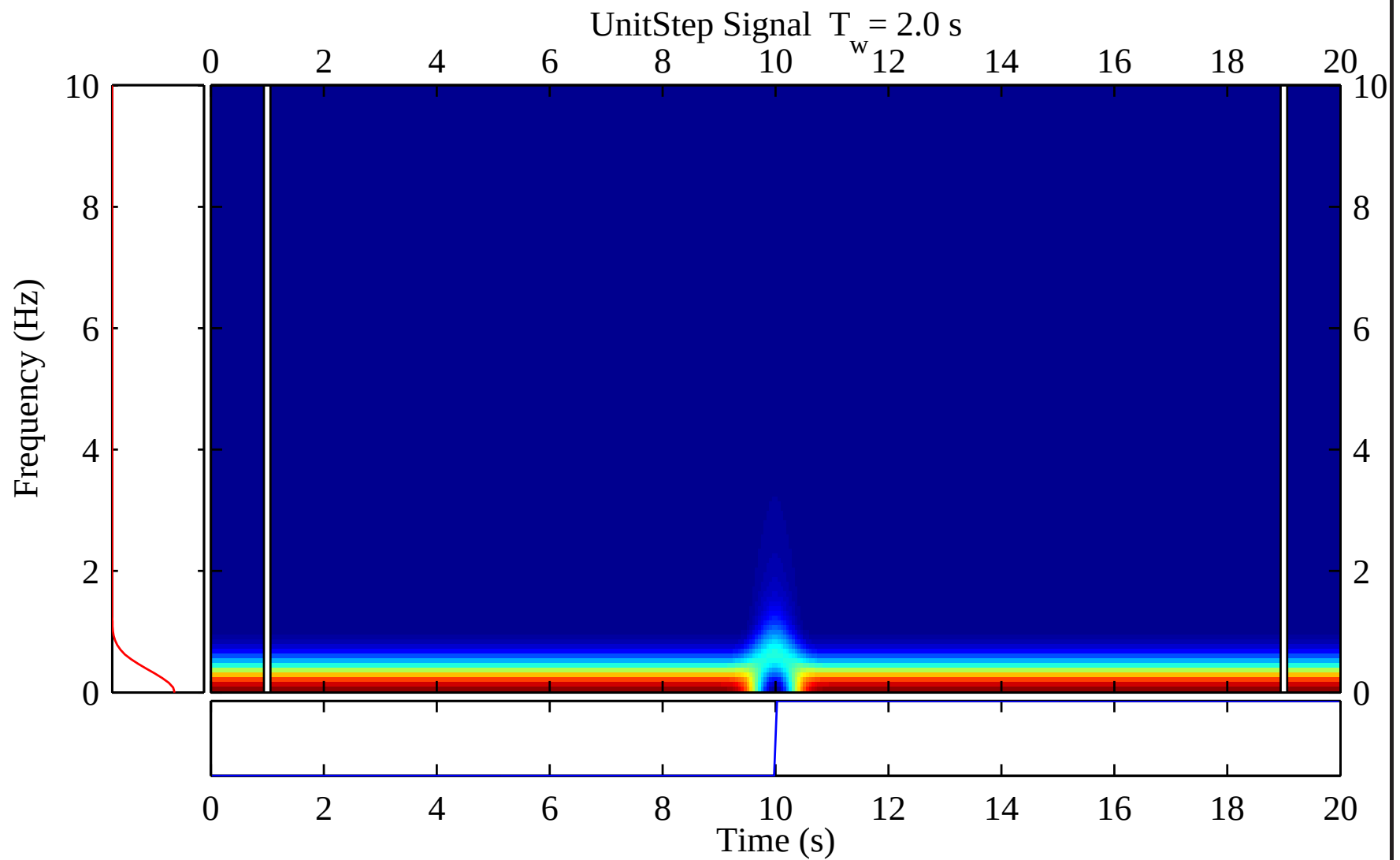
Example 2: Unit Step Spectrogram



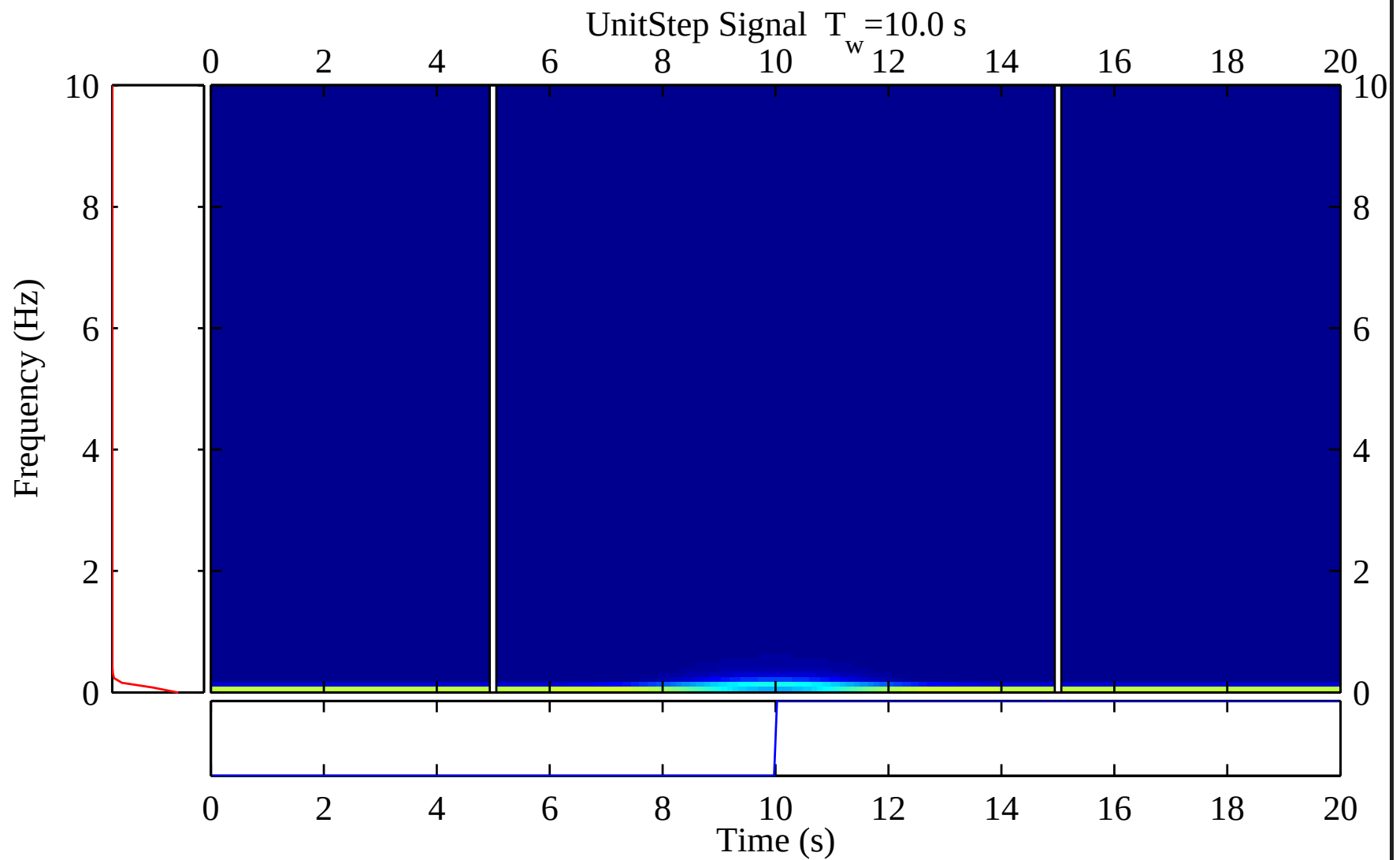
Example 2: Unit Step Spectrogram



Example 2: Unit Step Spectrogram



Example 2: Unit Step Spectrogram



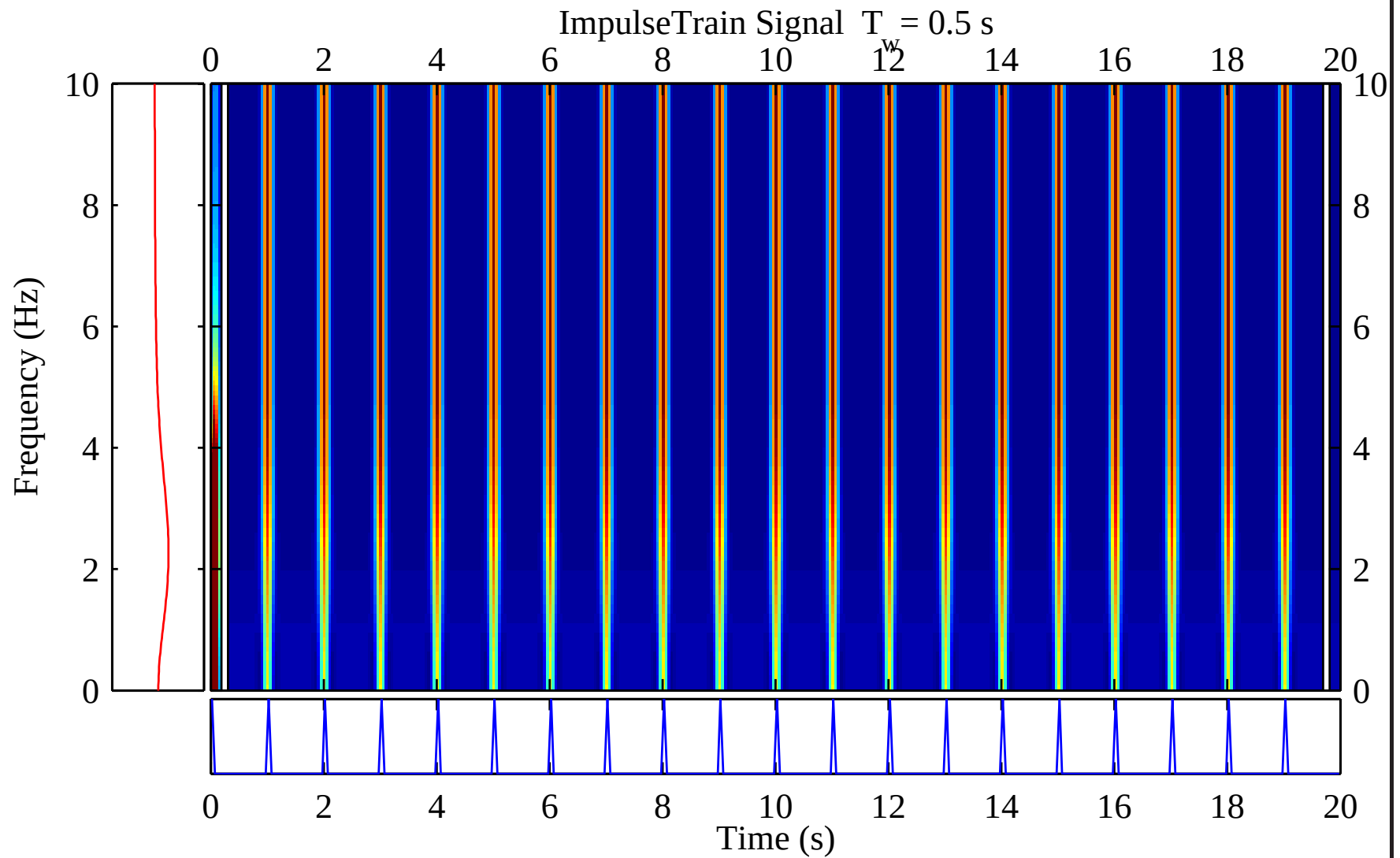
Periodic Signals

- Periodic signals consist of frequency components only at multiples of the fundamental frequency
- If the window length is smaller than the fundamental period, these areas analyzed as discrete events (insufficient frequency resolution)
- If the window length is several times the fundamental period, these areas appear to be periodic

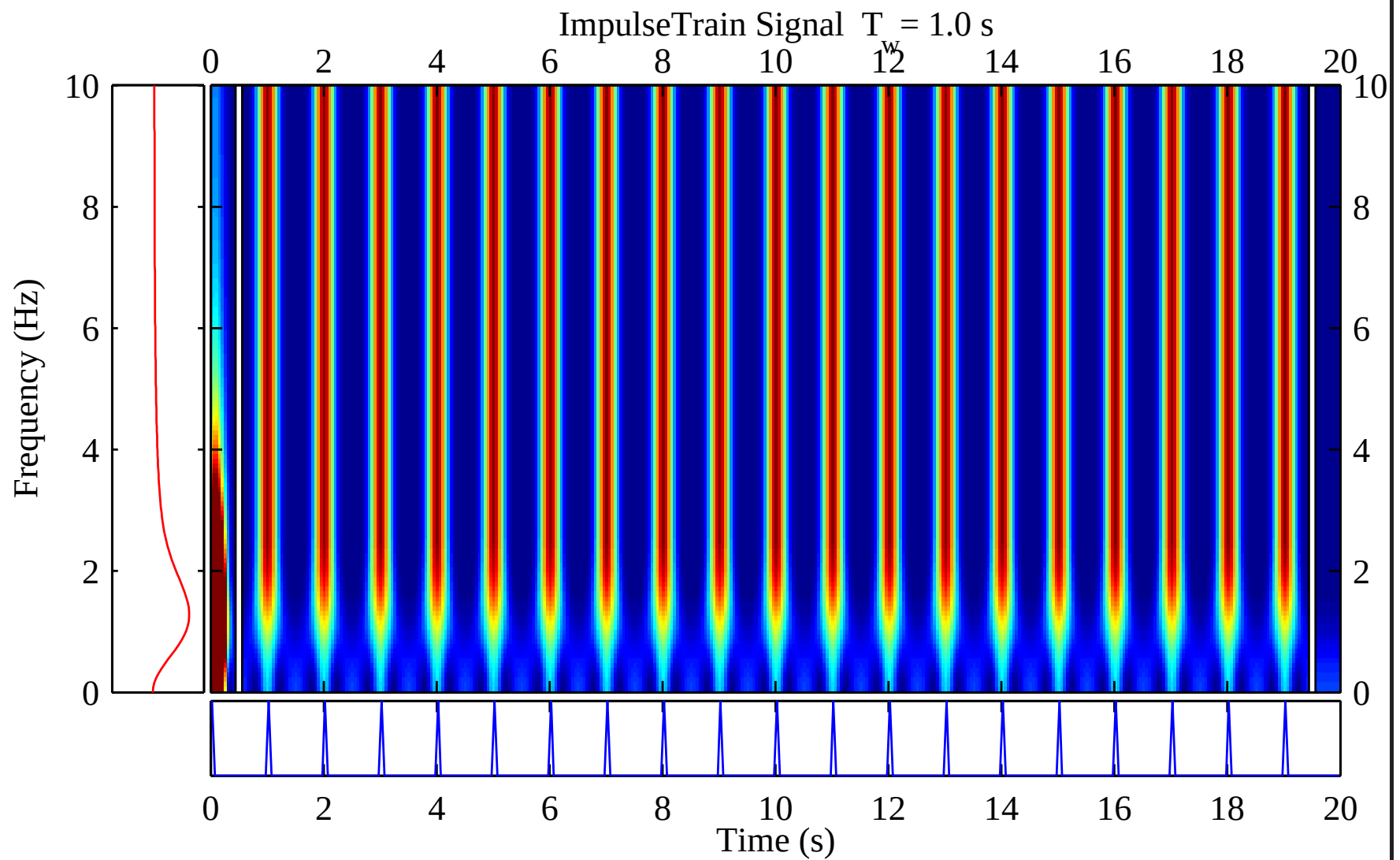
Example 3: Impulse Train Spectrograms

Repeat the previous example for a 1 Hz impulse train.

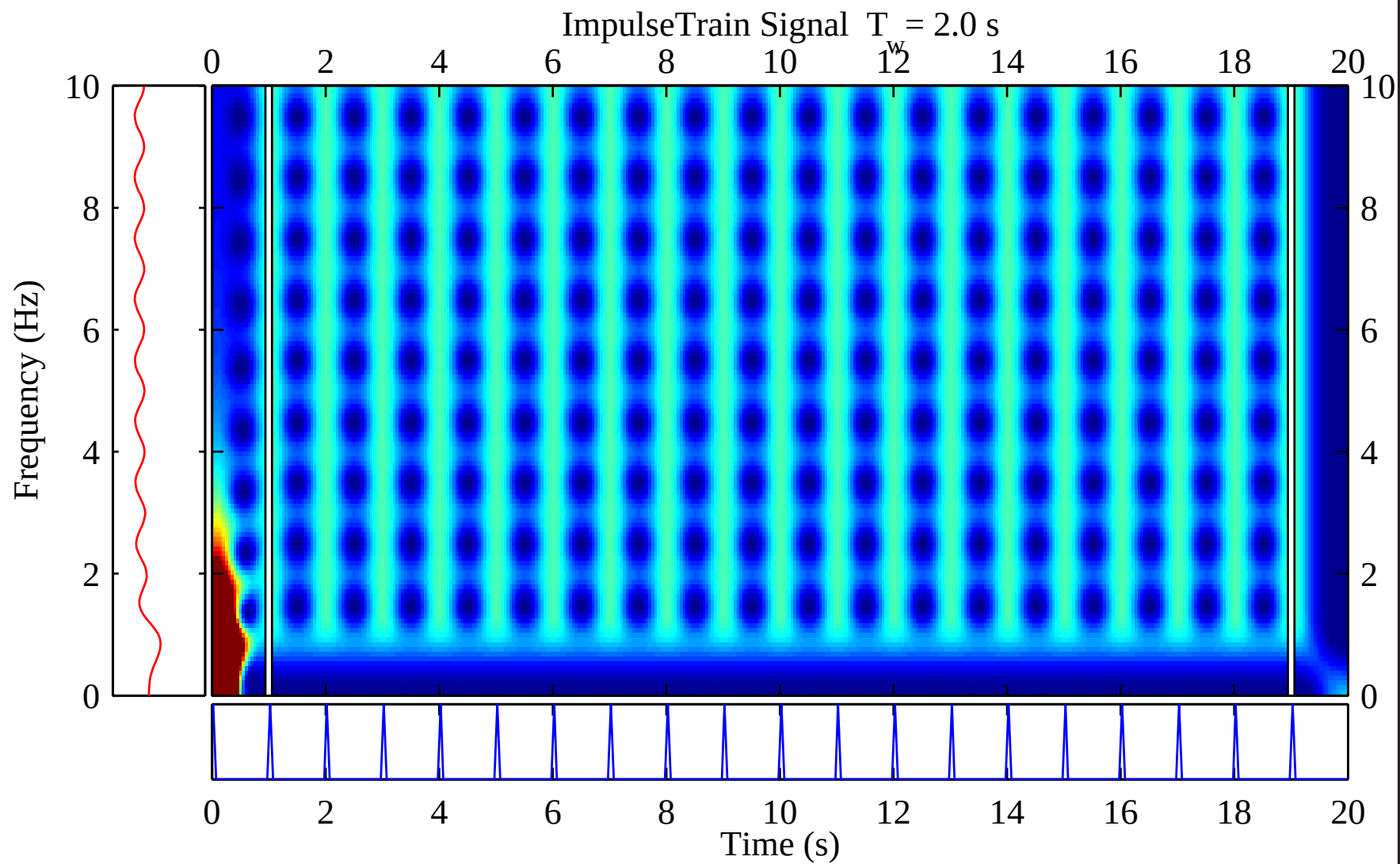
Example 3: Impulse Train Spectrogram



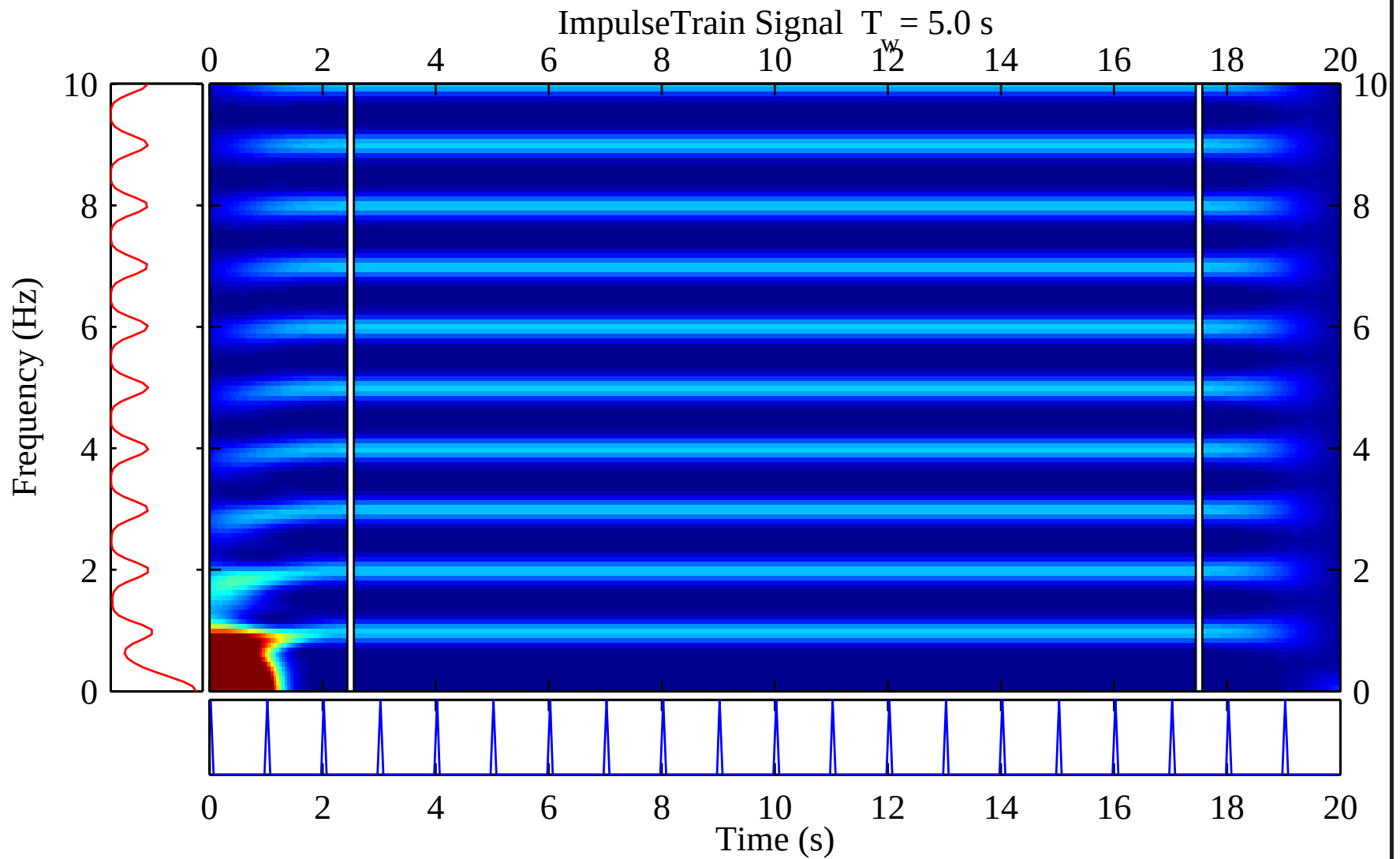
Example 3: Impulse Train Spectrogram



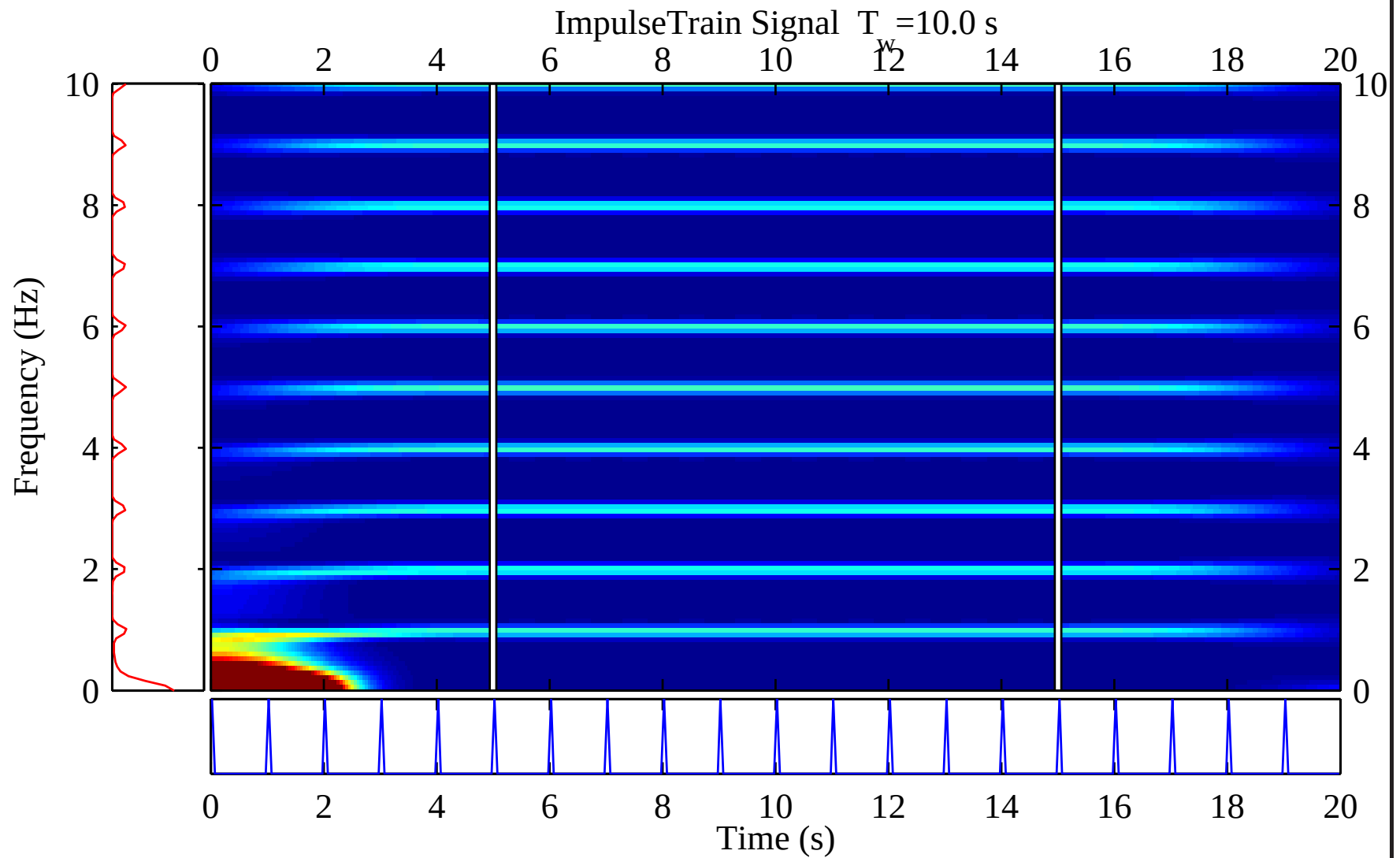
Example 3: Impulse Train Spectrogram



Example 3: Impulse Train Spectrogram



Example 3: Impulse Train Spectrogram



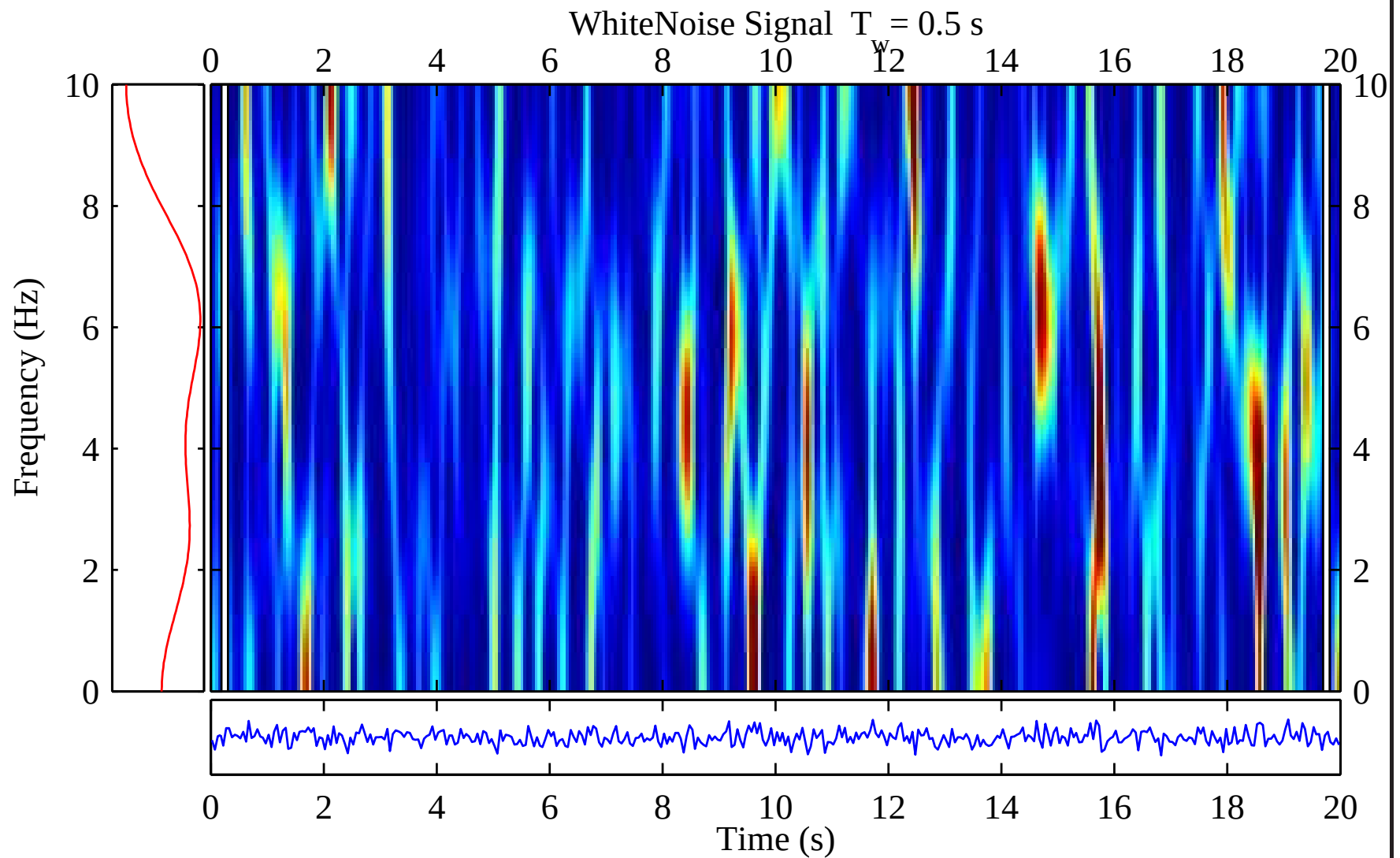
Random Signals

- It's important to be able to distinguish significant features from random fluctuations due to estimation error
- Recall that the periodogram had excessive variance
- Helpful to examine the spectrograms of white noise

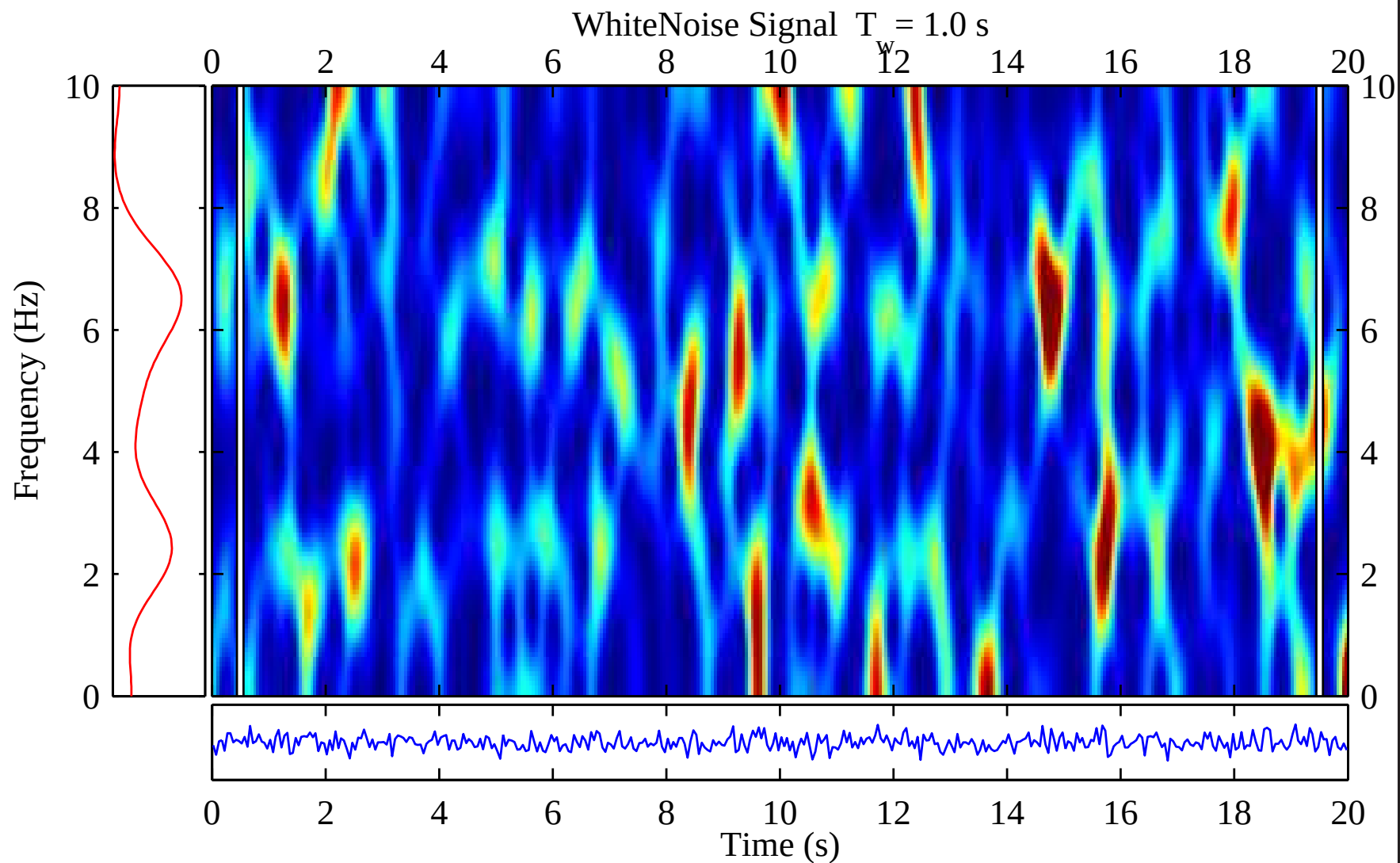
Example 4: White Noise Spectrograms

Repeat the previous example for a white noise signal.

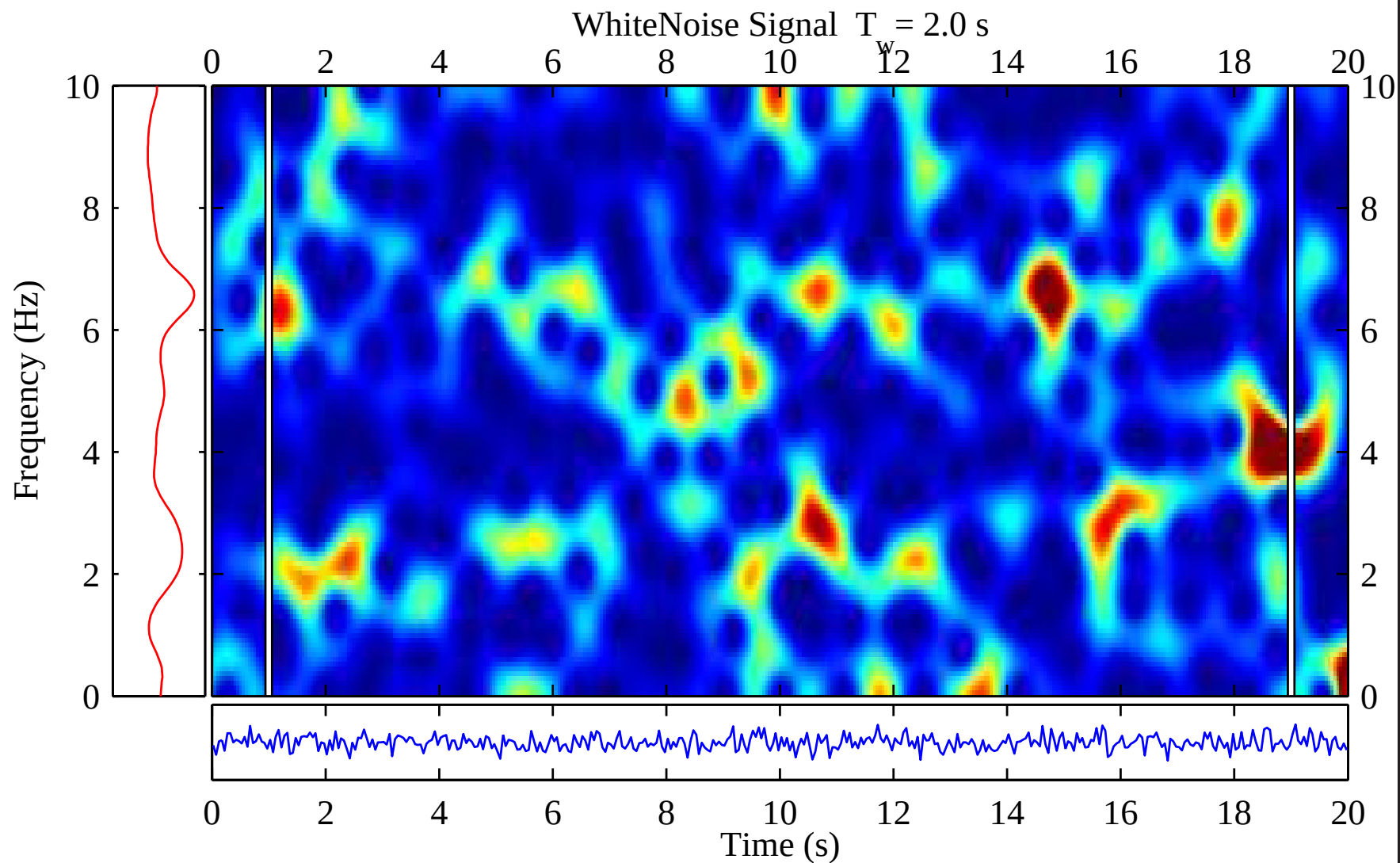
Example 4: White Noise Spectrogram



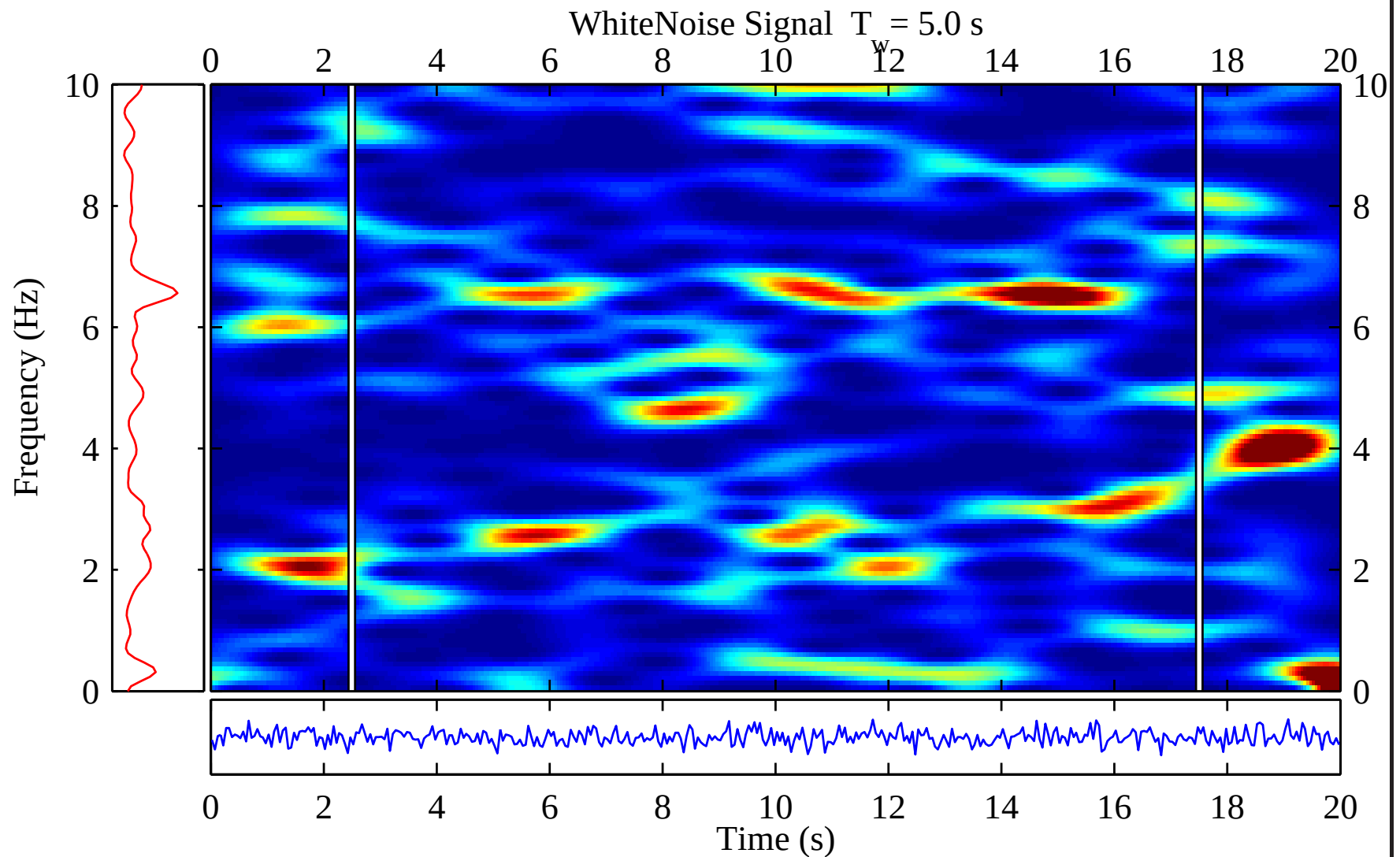
Example 4: White Noise Spectrogram



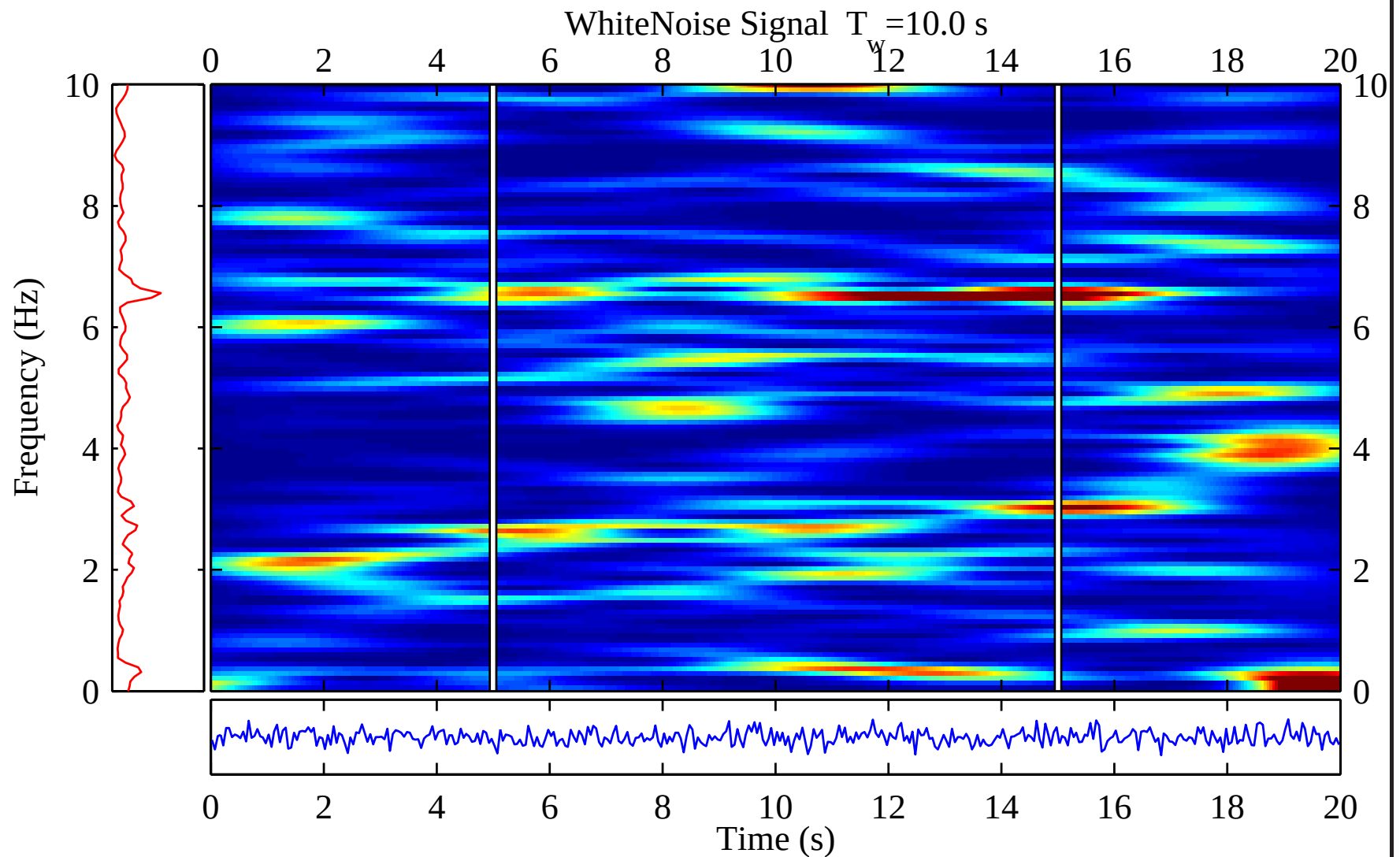
Example 4: White Noise Spectrogram



Example 4: White Noise Spectrogram



Example 4: White Noise Spectrogram



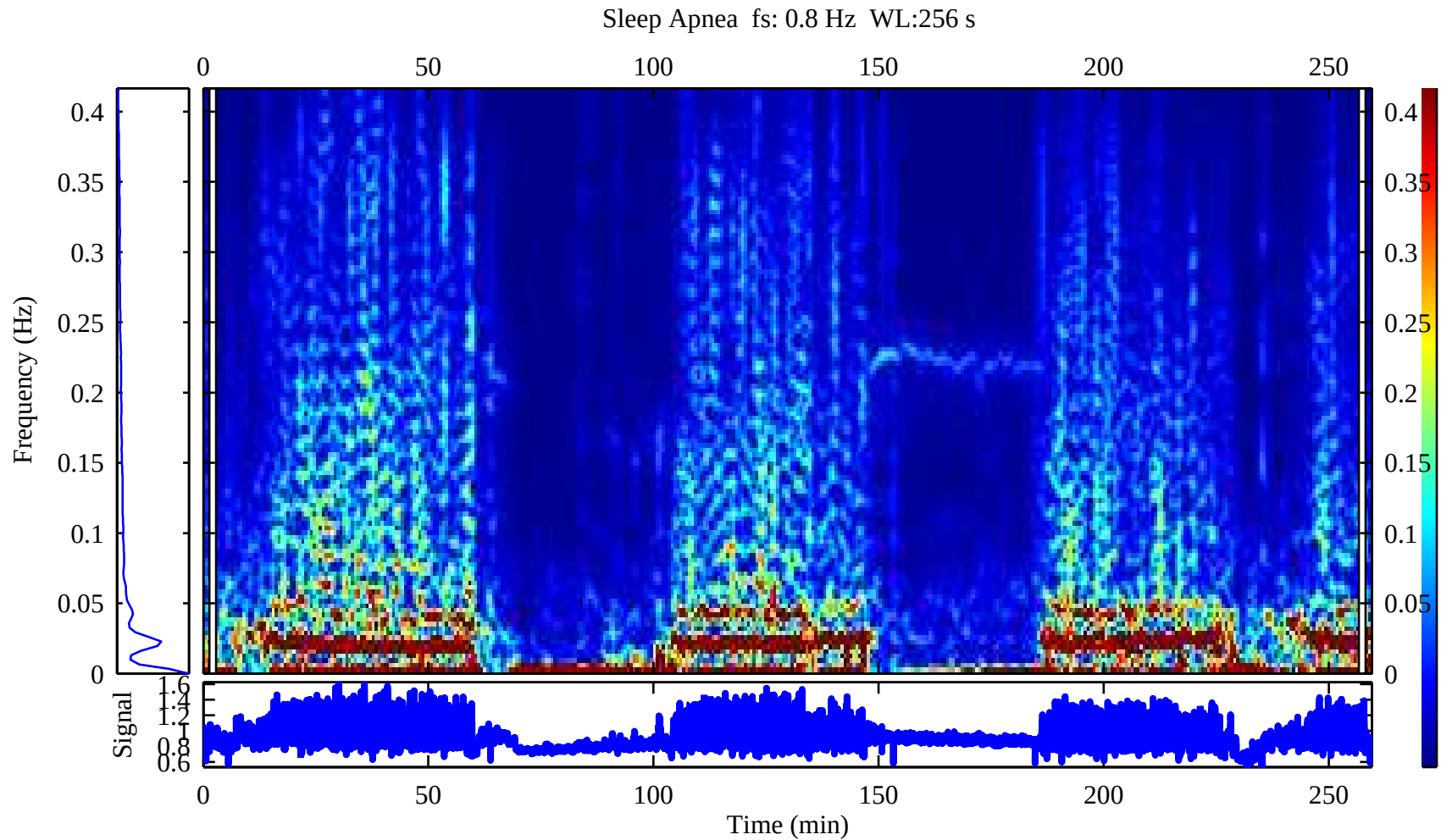
Example 5: Detecting Sleep Apnea

- Can obstructive sleep apnea be detected from the electrocardiogram alone?
- If possible, could build simple and cheap, portable diagnostic device
- International competition in 2000
- Data set
 - 25 recordings labelled by expert
 - 10 unlabelled recording
 - Approximately 8 hours each
- Objective: detect OSA for every minute of the 10 recordings

Example 5: Detecting Sleep Apnea

- Our approach was based on spectrograms
- Implemented QRS detector
- Applied spectrogram to instantaneous heart rate and other features
 - Interbeat intervals (R-R)
 - S amplitude
 - Pulse energy
- Winning entry!
 - 92.6% of minutes classified “correctly”
 - Third attempt

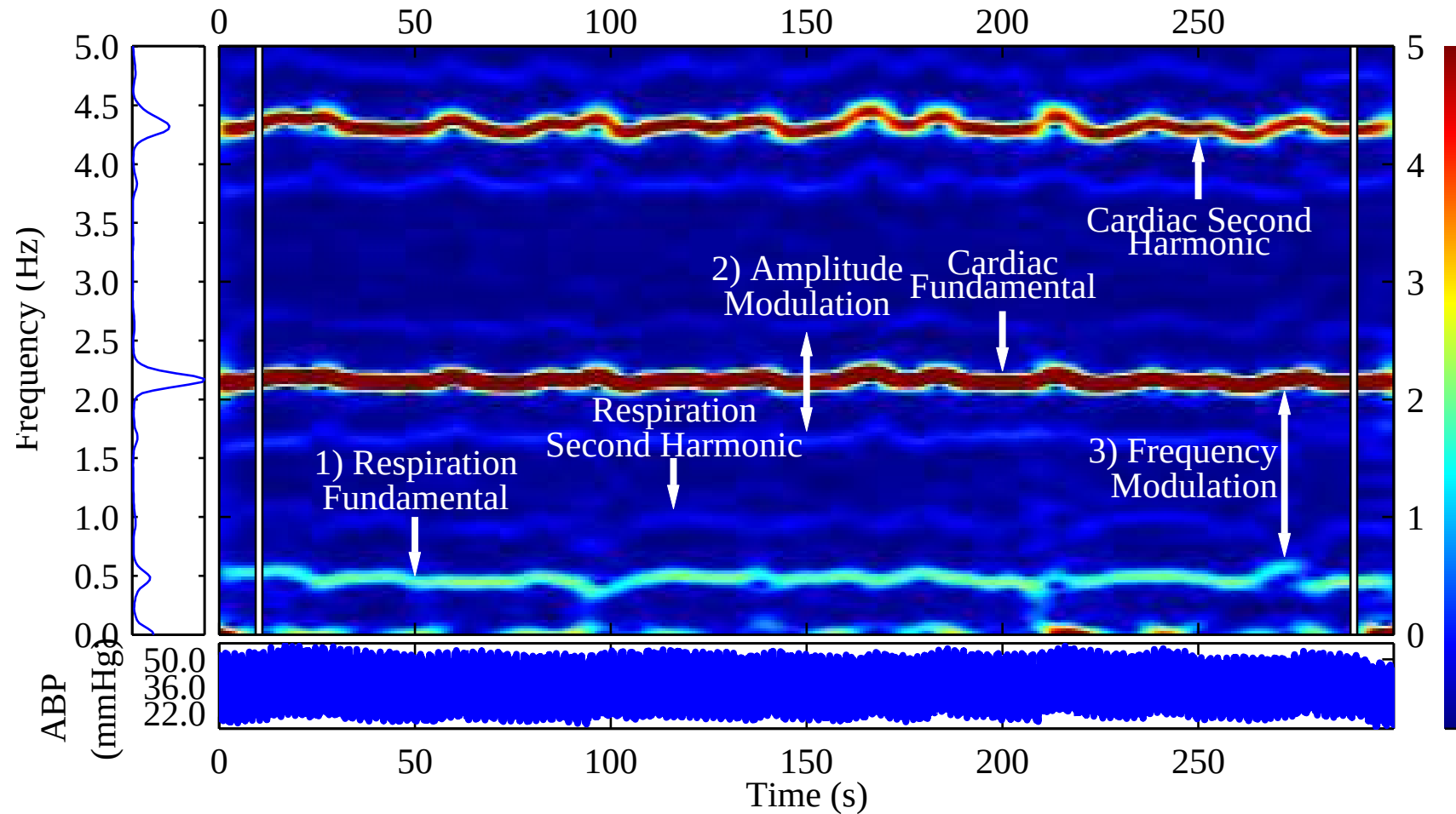
Example 5: Apnea Detection



Example 6: Arterial Blood Pressure

Generate the spectrogram of an arterial blood pressure signal from a child in an intensive care unit (ICU) with sepsis. Interpret the figure.

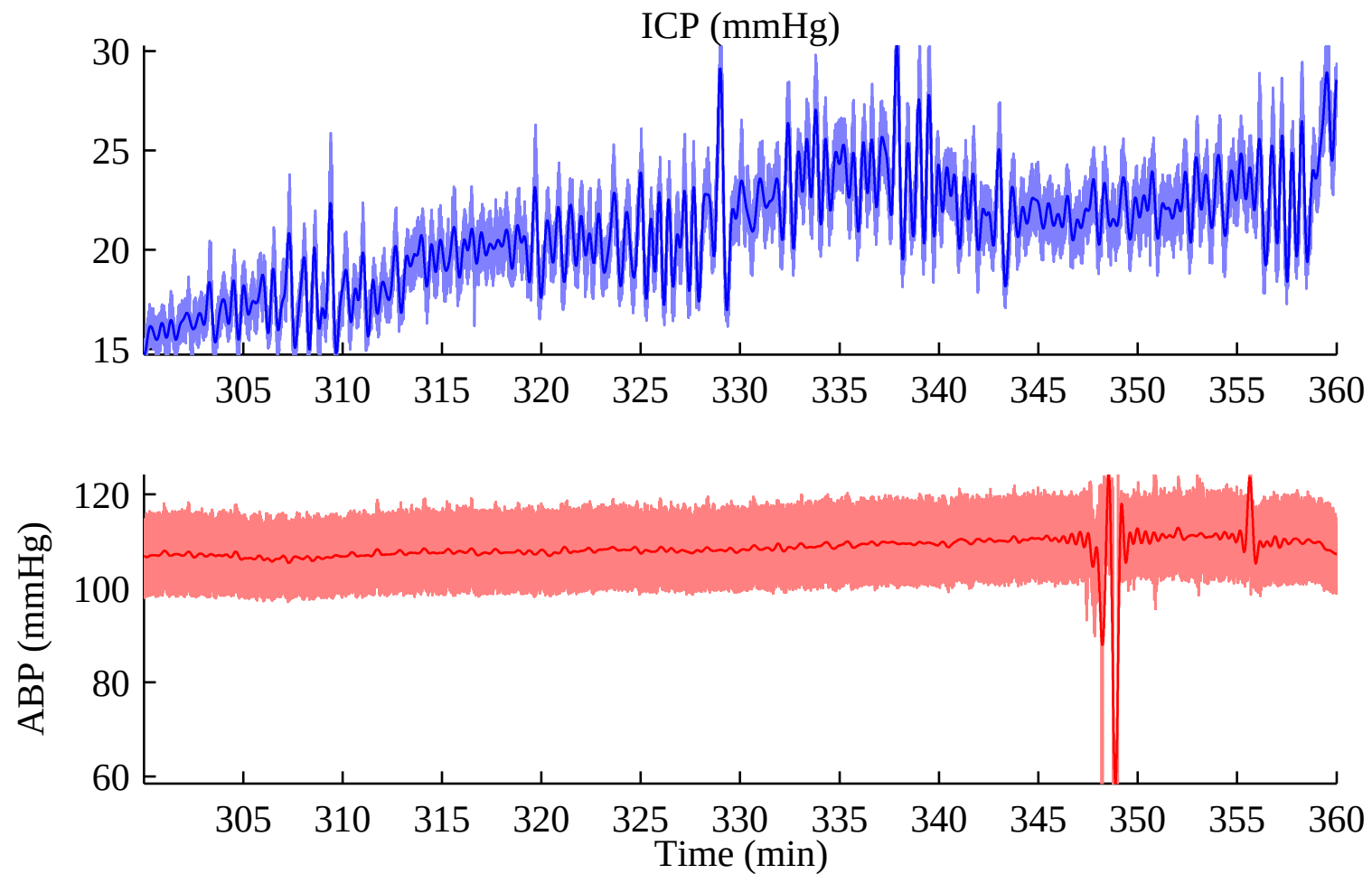
Example 6: ABP Spectrogram



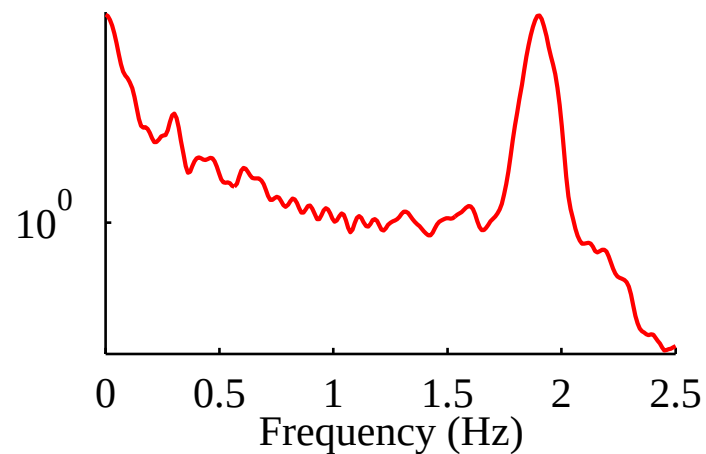
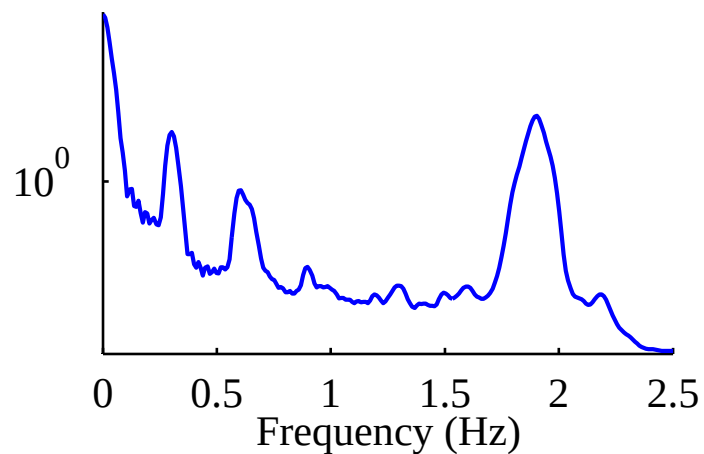
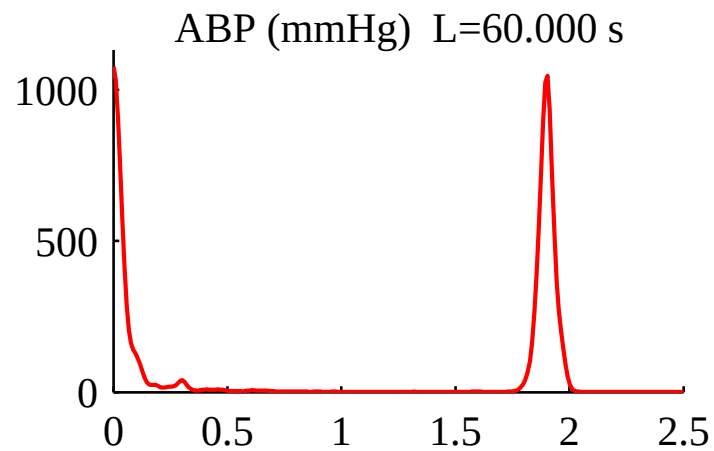
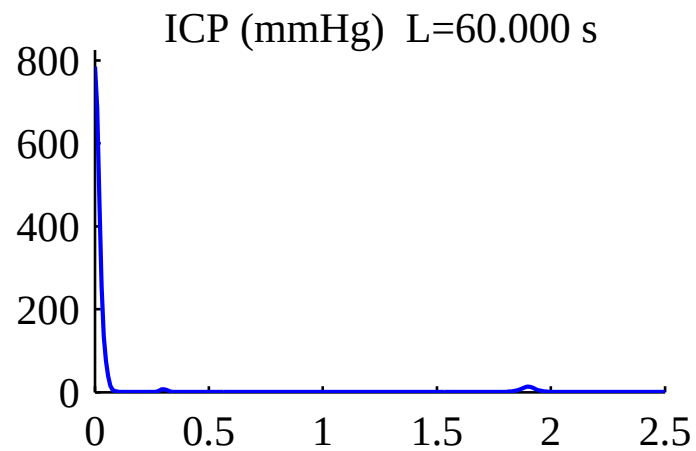
Example 7: ABP and ICP

Generate the coherenceogram of arterial blood pressure and intracranial pressure signals from a child with traumatic brain injury.

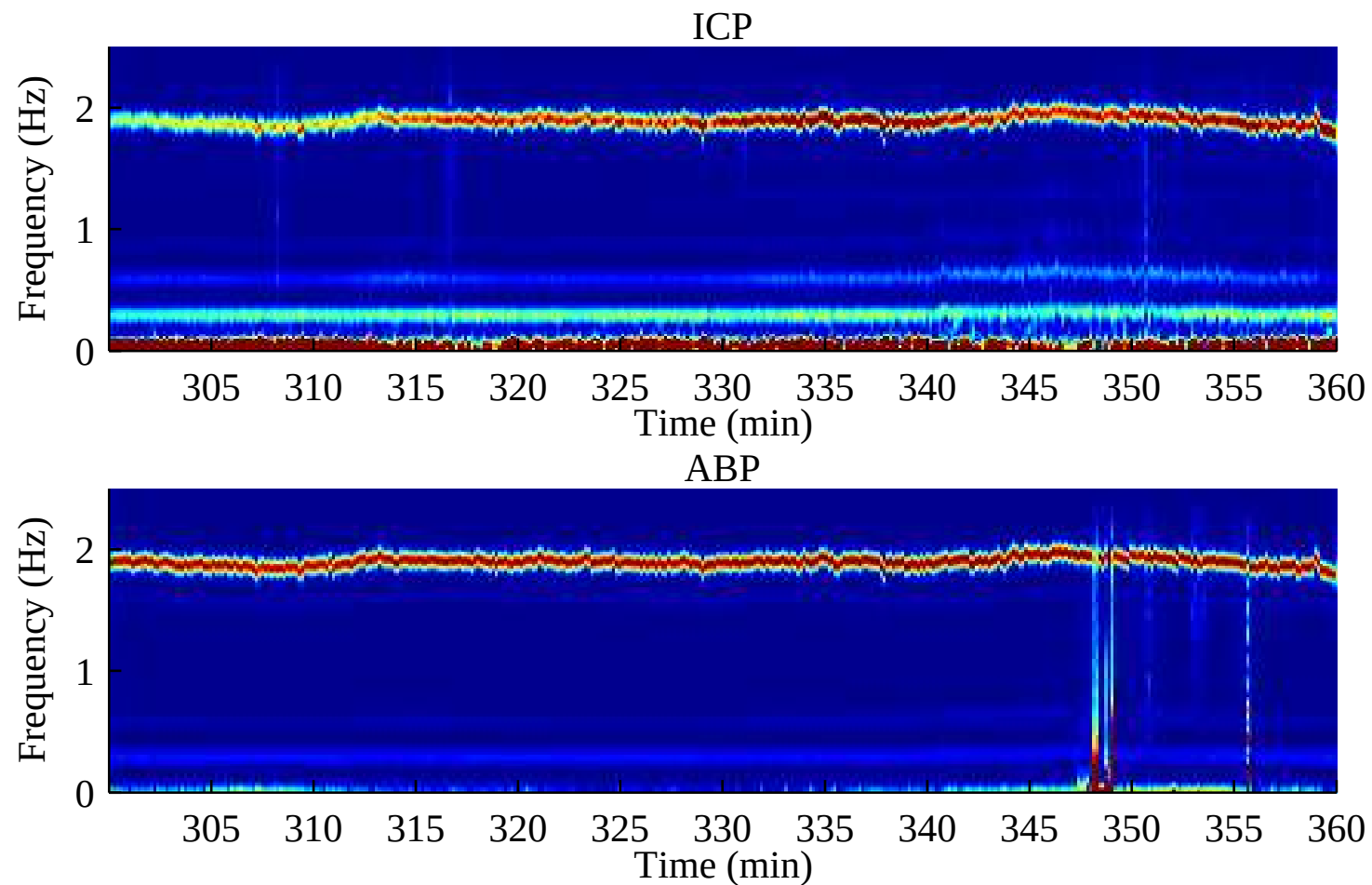
Example 7: ABP and ICP



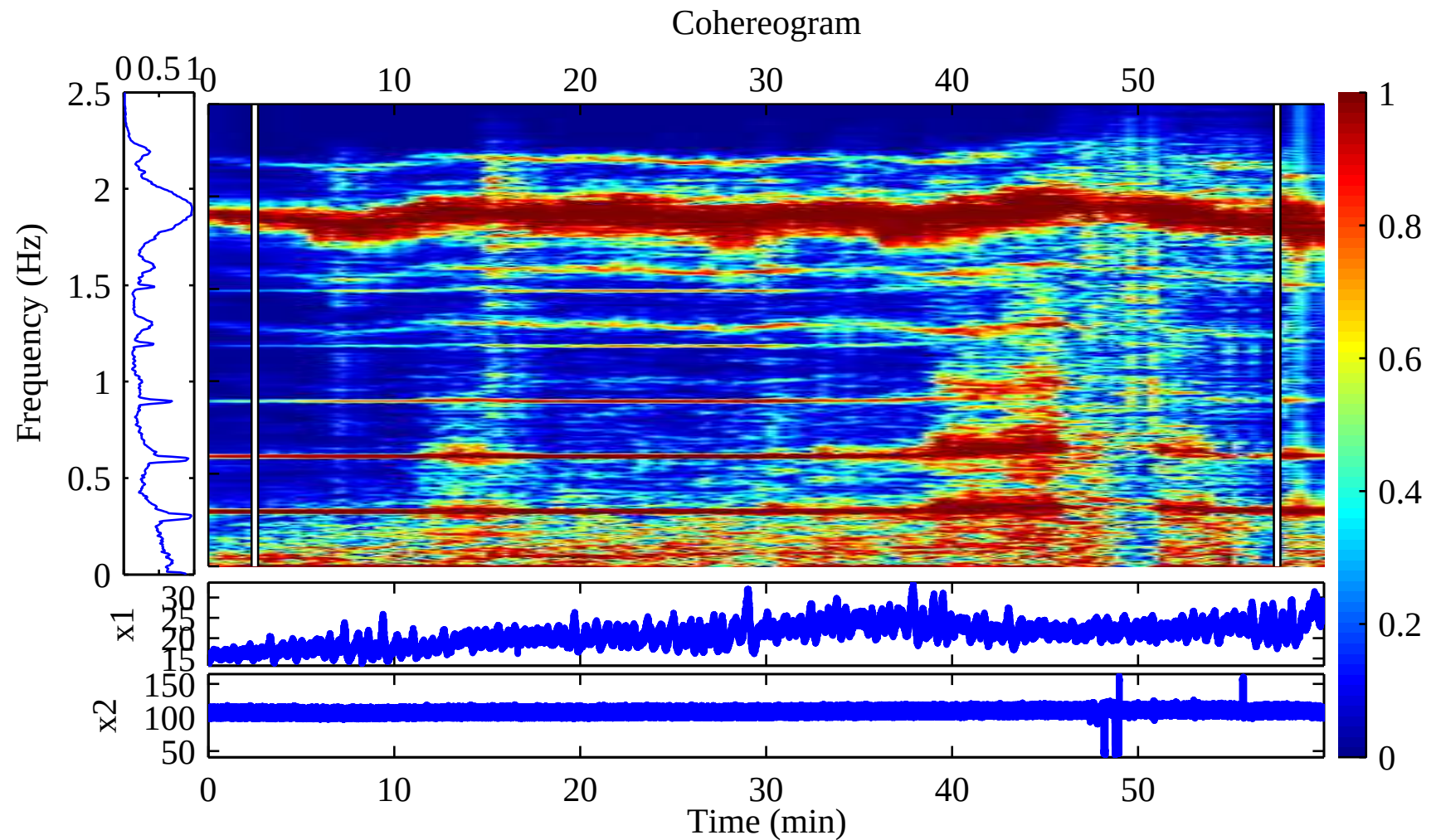
Example 7: ABP and ICP



Example 7: ABP and ICP



Example 7: ABP and ICP



Other Estimators

- Time-frequency analysis is a topic that warrants an entire class itself
- There are many estimators of how power is distributed across time and frequency
- Includes wavelets (“scaleograms”)

Summary

- Moving window estimates can be used to perform time-frequency analysis of locally stationary signals
- Conceptually, is the output of a bank of overlapping bandpass filters
- Window length is the most critical parameter
 - Controls tradeoff between time and frequency resolution
 - Duration-bandwidth product is limited by the uncertainty principle
- Can apply to joint estimation techniques as well
- Very useful for observing time-varying statistical properties

References